

Comprehensive Model Analysis Report

Generated on: 2026-02-11 14:41:18

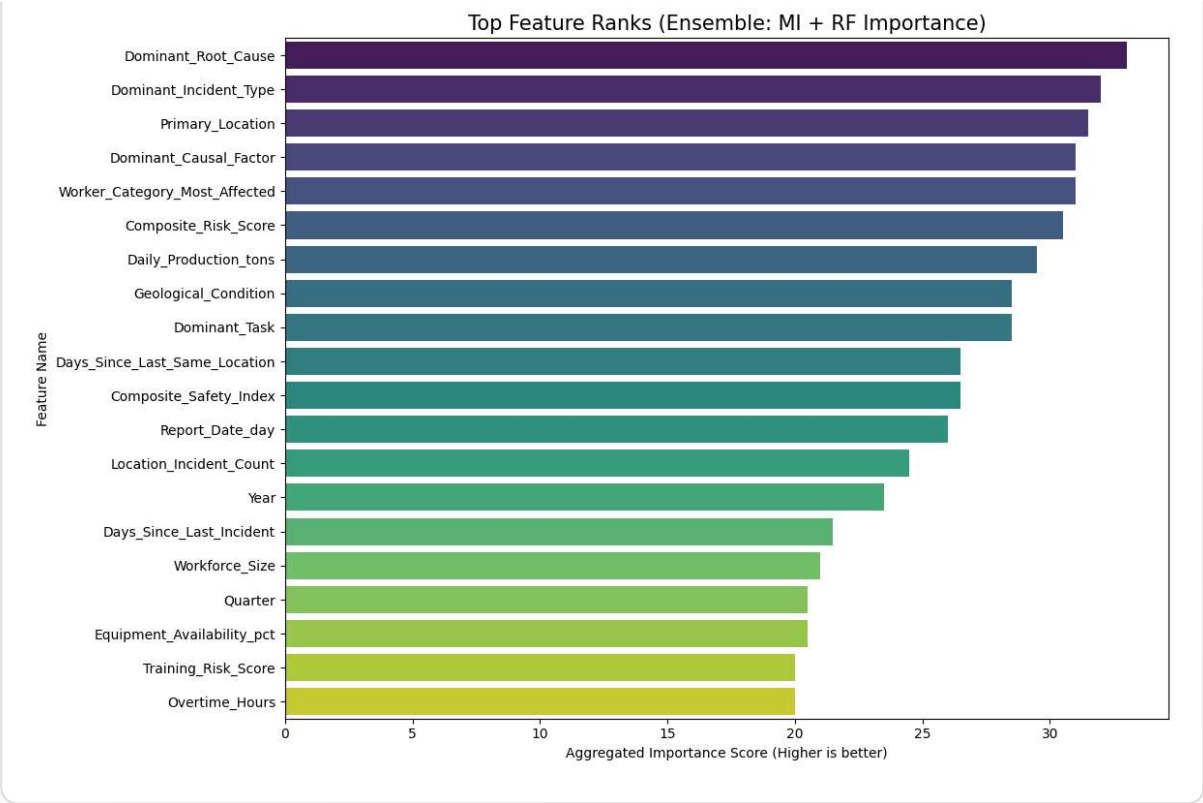
 Ensemble Feature Selection Active

Optimized Top 15 Features (Mutual Info + RF)

Selected: Dominant_Root_Cause, Dominant_Incident_Type, Primary_Location, Dominant_Causal_Factor, Worker_Category_Most_Affected, Composite_Risk_Score, Daily_Production_tons, Geological_Condition, Dominant_Task, Days_Since_Last_Same_Location, Composite_Safety_Index, Report_Date_day, Location_Incident_Count, Year, Days_Since_Last_Incident

0. Feature Selection Insights

Ensemble Feature Ranks



Top Selected Feature Scores

Feature Name	Combined Rank (Lower is Better)	MI Rank	RF Rank
Dominant_Root_Cause	7.5	1.0	14.0
Dominant_Incident_Type	8.5	4.0	13.0
Primary_Location	9.0	3.0	15.0
Dominant_Causal_Factor	9.5	9.0	10.0
Worker_Category_Most_Affected	9.5	7.0	12.0
Composite_Risk_Score	10.0	14.0	6.0
Daily_Production_tons	11.0	21.0	1.0
Geological_Condition	12.0	5.0	19.0
Dominant_Task	12.0	2.0	22.0

Feature Name	Combined Rank (Lower is Better)	MI Rank	RF Rank
Days_Since_Last_Same_Location	14.0	20.0	8.0
Composite_Safety_Index	14.0	19.0	9.0
Report_Date_day	14.5	22.0	7.0
Location_Incident_Count	16.0	8.0	24.0
Year	17.0	16.0	18.0
Days_Since_Last_Incident	19.0	36.0	2.0
Workforce_Size	19.5	36.0	3.0
Quarter	20.0	10.0	30.0
Equipment_Availability_pct	20.0	36.0	4.0
Training_Risk_Score	20.5	6.0	35.0
Overtime_Hours	20.5	36.0	5.0

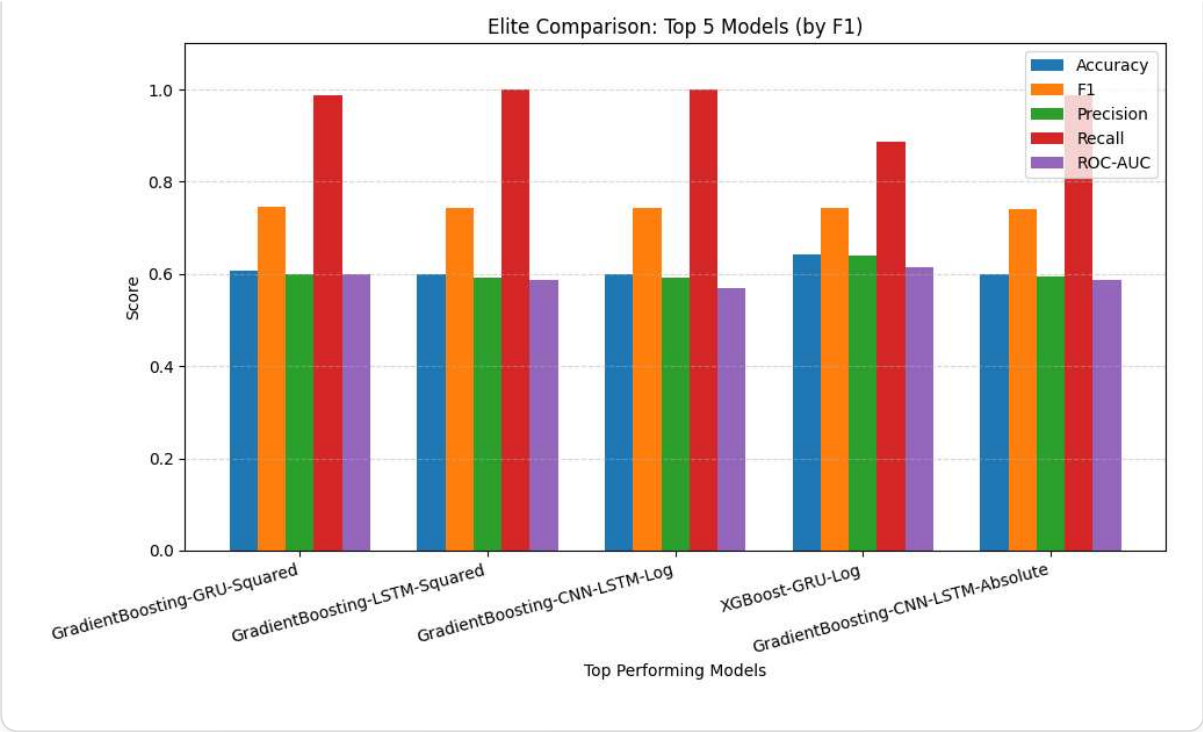


Top Recommendation: GradientBoosting-GRU-Squared

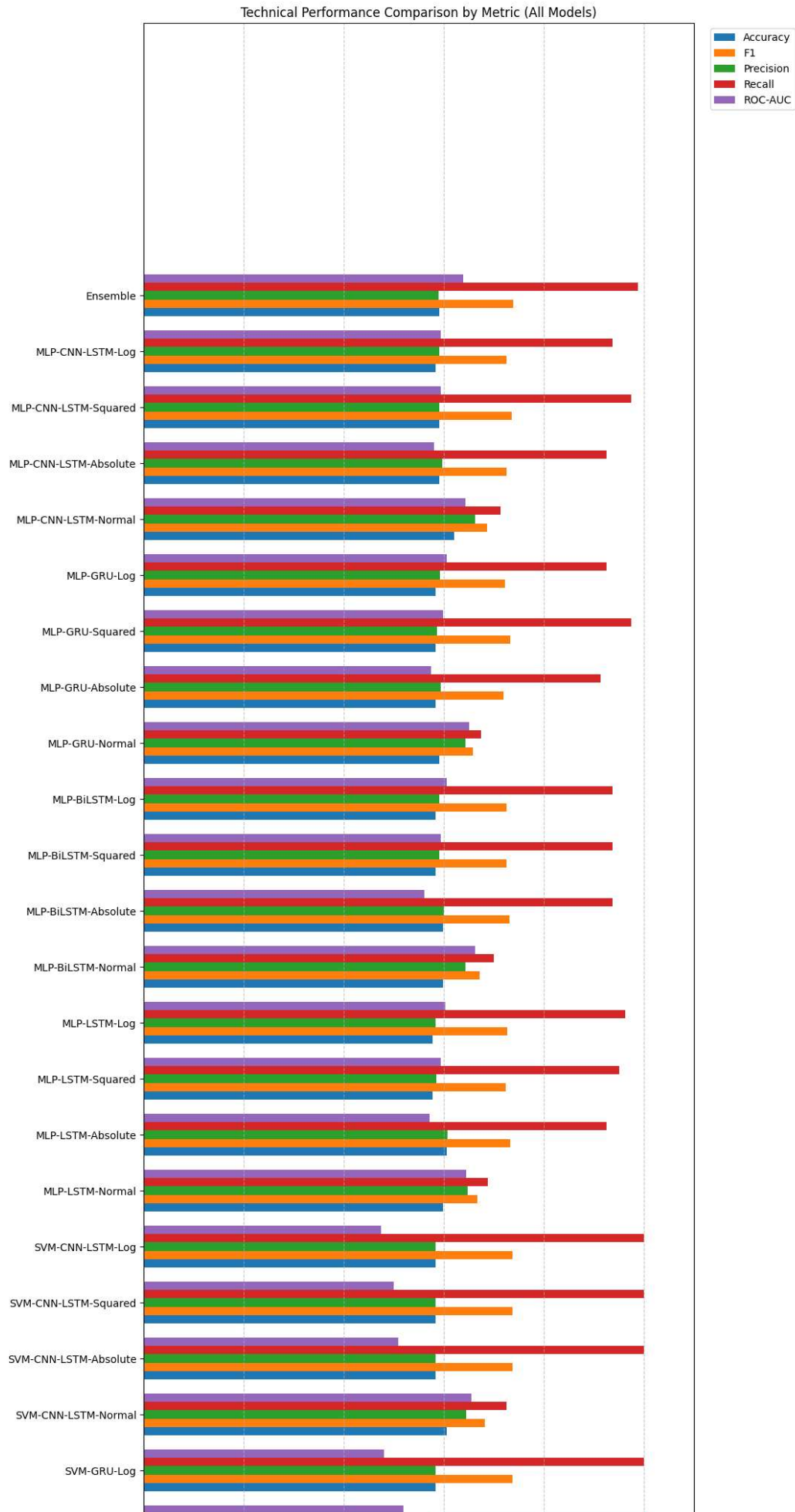
Highest F1 Score achieving optimal balance between precision and recall.

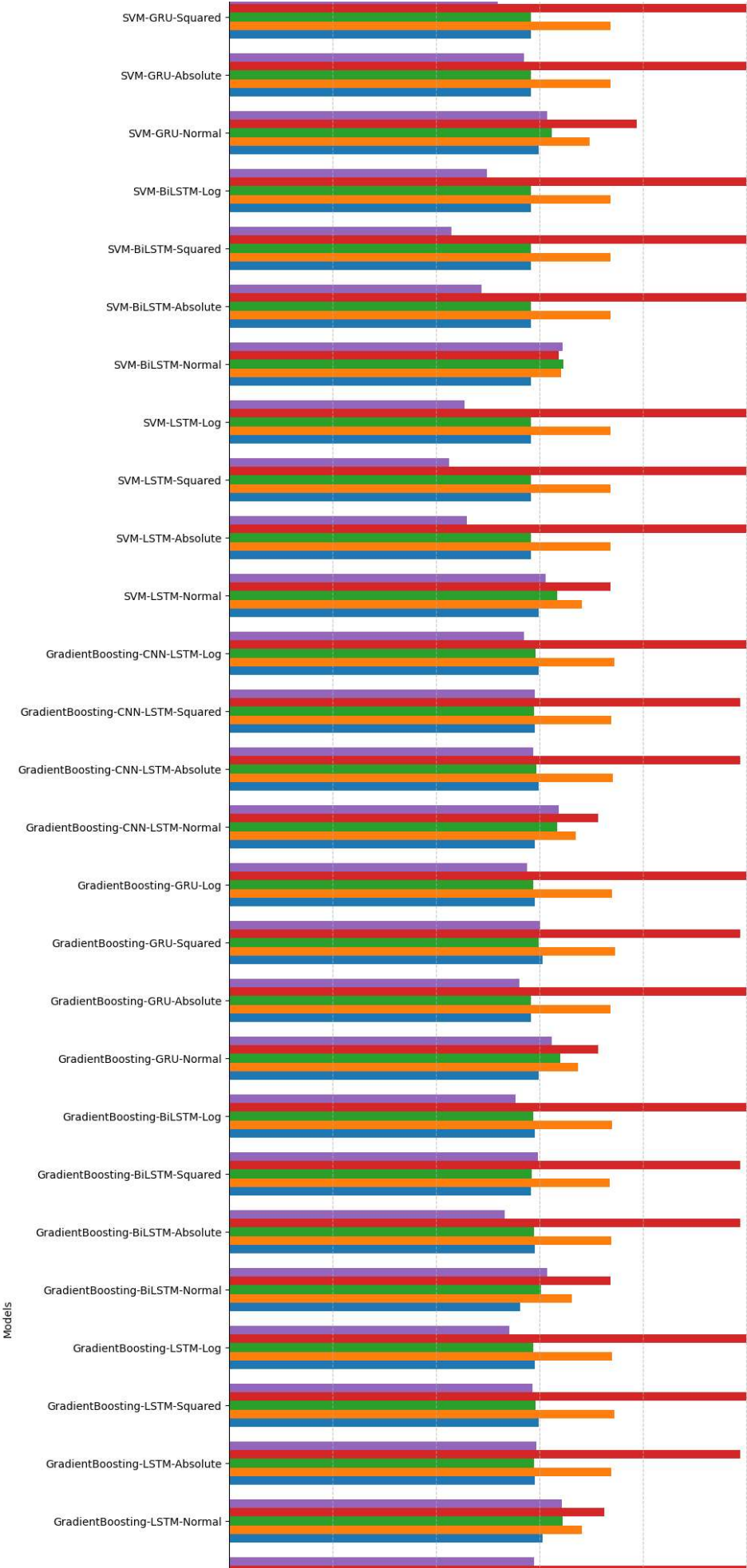
1. Global Performance Summaries

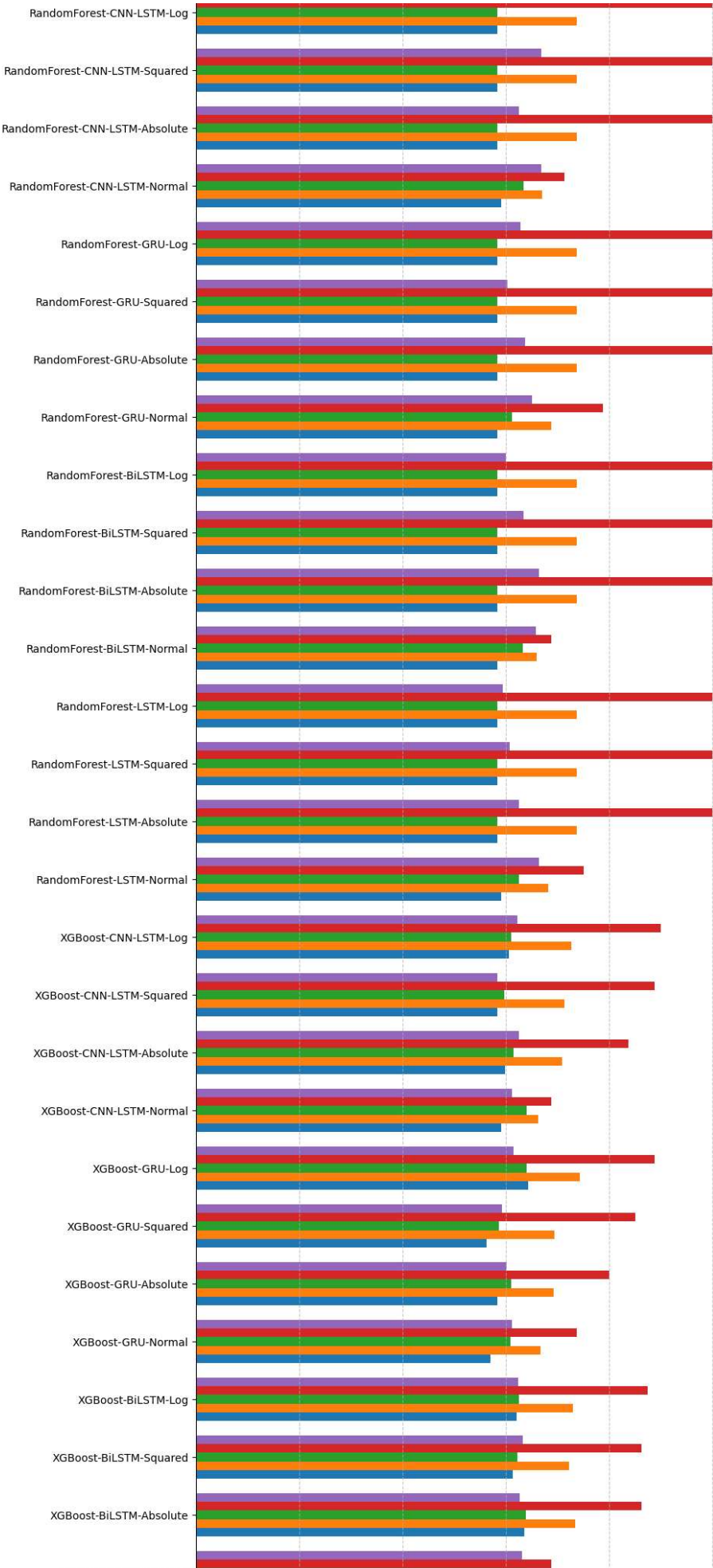
Elite Performance (Top 5)

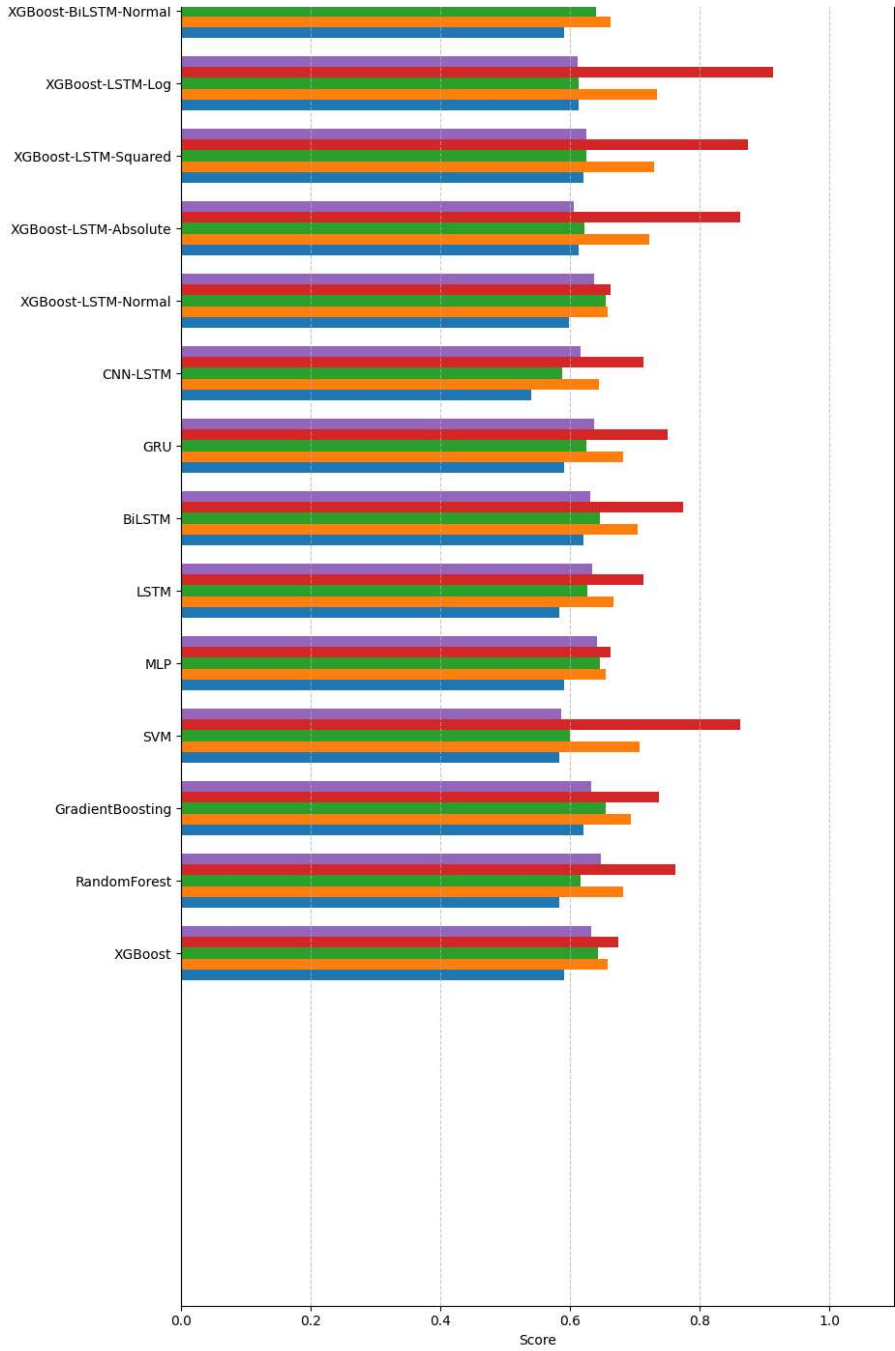


Metric Comparison (All Models)

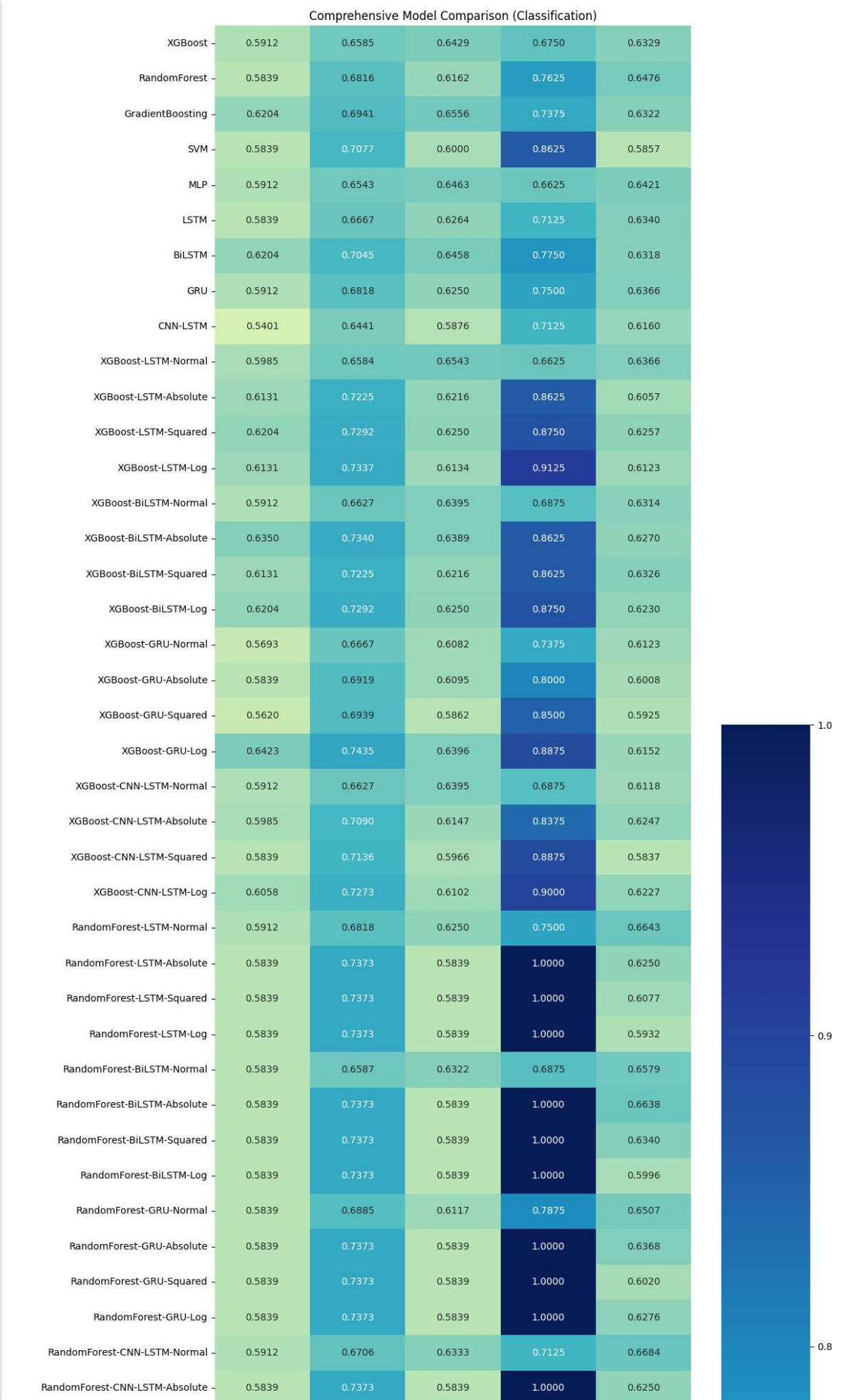


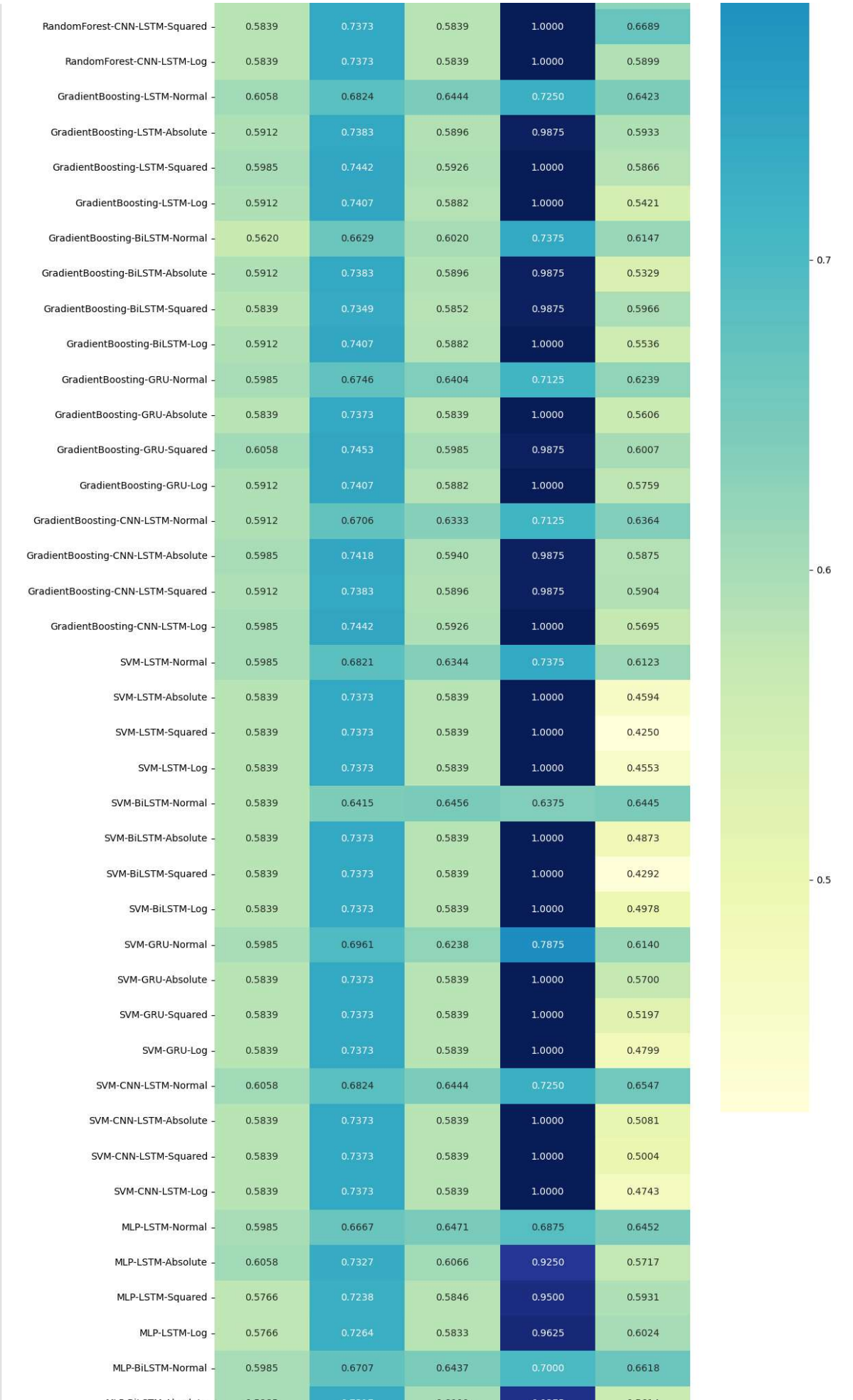






Classification Relationship Matrix

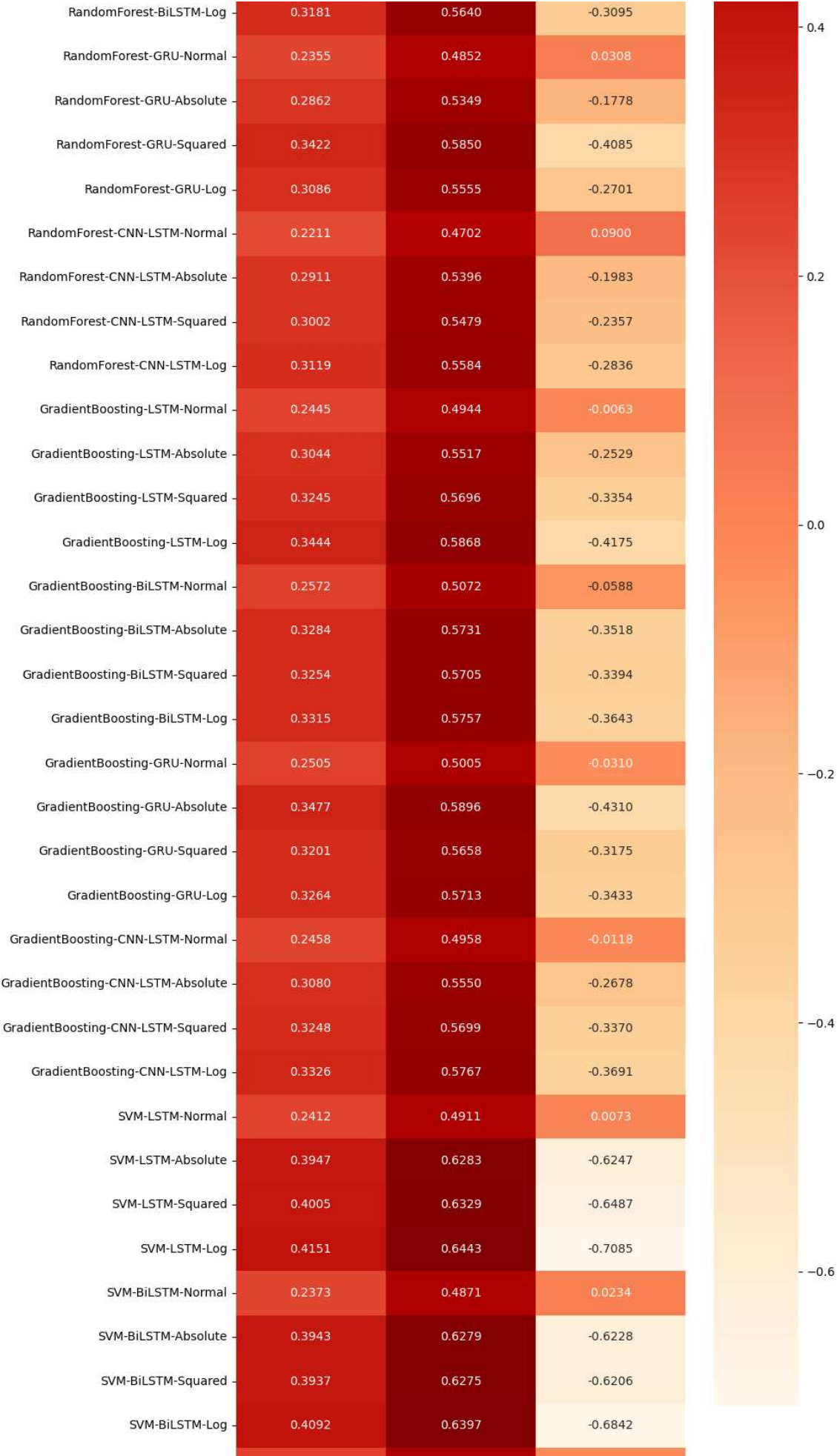


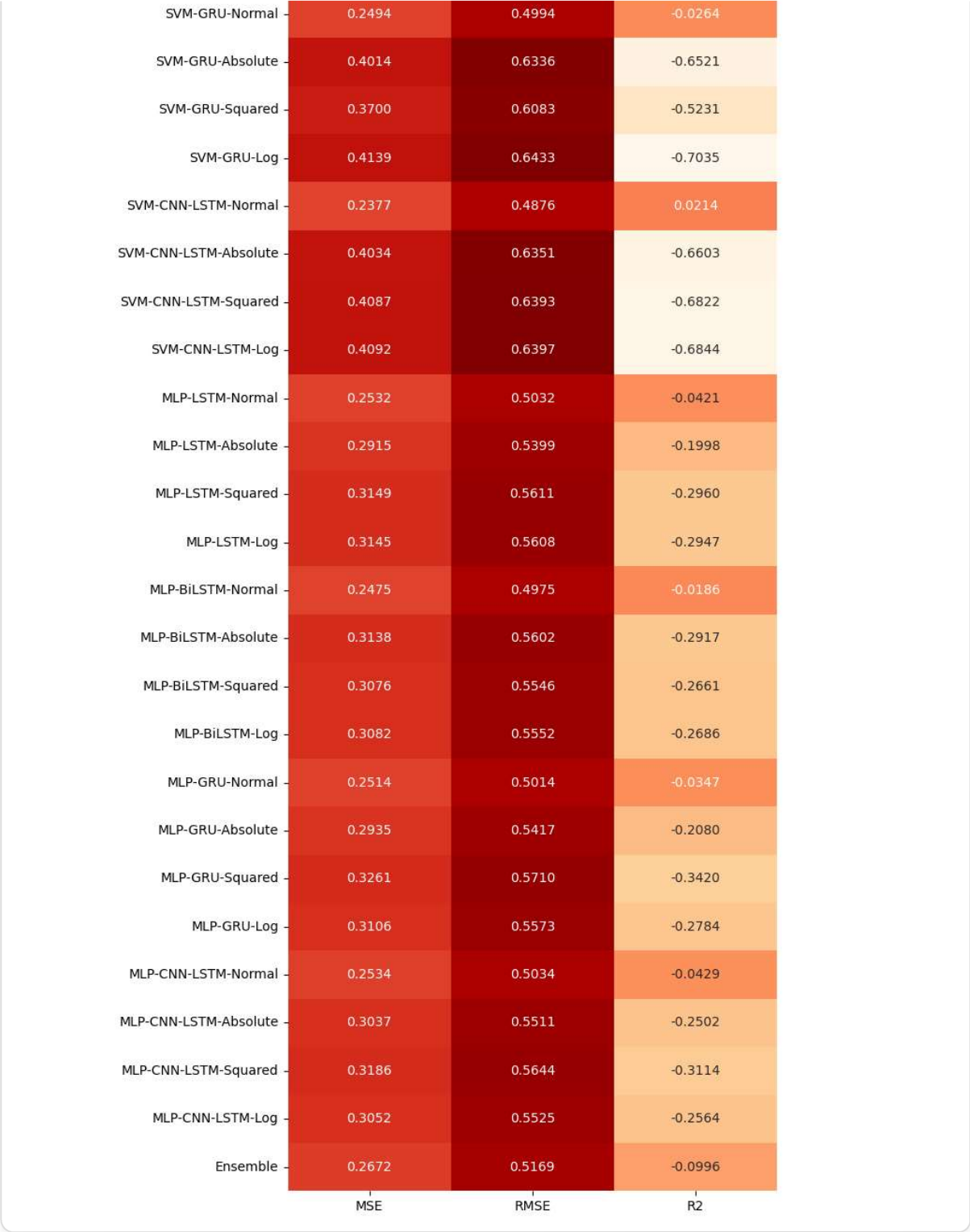


MLP-BiLSTM-Absolute	0.5985	0.7317	0.6000	0.9375	0.5614
MLP-BiLSTM-Squared	0.5839	0.7246	0.5906	0.9375	0.5940
MLP-BiLSTM-Log	0.5839	0.7246	0.5906	0.9375	0.6061
MLP-GRU-Normal	0.5912	0.6585	0.6429	0.6750	0.6509
MLP-GRU-Absolute	0.5839	0.7192	0.5935	0.9125	0.5738
MLP-GRU-Squared	0.5839	0.7324	0.5865	0.9750	0.5976
MLP-GRU-Log	0.5839	0.7220	0.5920	0.9250	0.6060
MLP-CNN-LSTM-Normal	0.6204	0.6867	0.6628	0.7125	0.6430
MLP-CNN-LSTM-Absolute	0.5912	0.7255	0.5968	0.9250	0.5800
MLP-CNN-LSTM-Squared	0.5912	0.7358	0.5909	0.9750	0.5942
MLP-CNN-LSTM-Log	0.5839	0.7246	0.5906	0.9375	0.5939
Ensemble	0.5912	0.7383	0.5896	0.9875	0.6384
	Accuracy	F1	Precision	Recall	ROC-AUC

Regression Error Matrix







2. Consolidated Metrics Table

Model	Accuracy	F1	Precision	Recall	ROC-AUC
XGBoost	0.5912	0.6585	0.6429	0.6750	0.6329

Model	Accuracy	F1	Precision	Recall	ROC-AUC
RandomForest	0.5839	0.6816	0.6162	0.7625	0.6476
GradientBoosting	0.6204	0.6941	0.6556	0.7375	0.6322
SVM	0.5839	0.7077	0.6000	0.8625	0.5857
MLP	0.5912	0.6543	0.6463	0.6625	0.6421
LSTM	0.5839	0.6667	0.6264	0.7125	0.6340
BiLSTM	0.6204	0.7045	0.6458	0.7750	0.6318
GRU	0.5912	0.6818	0.6250	0.7500	0.6366
CNN-LSTM	0.5401	0.6441	0.5876	0.7125	0.6160
XGBoost-LSTM-Normal	0.5985	0.6584	0.6543	0.6625	0.6366
XGBoost-LSTM-Absolute	0.6131	0.7225	0.6216	0.8625	0.6057
XGBoost-LSTM-Squared	0.6204	0.7292	0.6250	0.8750	0.6257
XGBoost-LSTM-Log	0.6131	0.7337	0.6134	0.9125	0.6123
XGBoost-BiLSTM-Normal	0.5912	0.6627	0.6395	0.6875	0.6314
XGBoost-BiLSTM-Absolute	0.6350	0.7340	0.6389	0.8625	0.6270
XGBoost-BiLSTM-Squared	0.6131	0.7225	0.6216	0.8625	0.6326
XGBoost-BiLSTM-Log	0.6204	0.7292	0.6250	0.8750	0.6230
XGBoost-GRU-Normal	0.5693	0.6667	0.6082	0.7375	0.6123
XGBoost-GRU-Absolute	0.5839	0.6919	0.6095	0.8000	0.6008

Model	Accuracy	F1	Precision	Recall	ROC-AUC
XGBoost-GRU-Squared	0.5620	0.6939	0.5862	0.8500	0.5925
XGBoost-GRU-Log	0.6423	0.7435	0.6396	0.8875	0.6152
XGBoost-CNN-LSTM-Normal	0.5912	0.6627	0.6395	0.6875	0.6118
XGBoost-CNN-LSTM-Absolute	0.5985	0.7090	0.6147	0.8375	0.6247
XGBoost-CNN-LSTM-Squared	0.5839	0.7136	0.5966	0.8875	0.5837
XGBoost-CNN-LSTM-Log	0.6058	0.7273	0.6102	0.9000	0.6227
RandomForest-LSTM-Normal	0.5912	0.6818	0.6250	0.7500	0.6643
RandomForest-LSTM-Absolute	0.5839	0.7373	0.5839	1.0000	0.6250
RandomForest-LSTM-Squared	0.5839	0.7373	0.5839	1.0000	0.6077
RandomForest-LSTM-Log	0.5839	0.7373	0.5839	1.0000	0.5932
RandomForest-BiLSTM-Normal	0.5839	0.6587	0.6322	0.6875	0.6579
RandomForest-BiLSTM-Absolute	0.5839	0.7373	0.5839	1.0000	0.6638
RandomForest-BiLSTM-Squared	0.5839	0.7373	0.5839	1.0000	0.6340
RandomForest-BiLSTM-Log	0.5839	0.7373	0.5839	1.0000	0.5996

Model	Accuracy	F1	Precision	Recall	ROC-AUC
RandomForest-GRU-Normal	0.5839	0.6885	0.6117	0.7875	0.6507
RandomForest-GRU-Absolute	0.5839	0.7373	0.5839	1.0000	0.6368
RandomForest-GRU-Squared	0.5839	0.7373	0.5839	1.0000	0.6020
RandomForest-GRU-Log	0.5839	0.7373	0.5839	1.0000	0.6276
RandomForest-CNN-LSTM-Normal	0.5912	0.6706	0.6333	0.7125	0.6684
RandomForest-CNN-LSTM-Absolute	0.5839	0.7373	0.5839	1.0000	0.6250
RandomForest-CNN-LSTM-Squared	0.5839	0.7373	0.5839	1.0000	0.6689
RandomForest-CNN-LSTM-Log	0.5839	0.7373	0.5839	1.0000	0.5899
GradientBoosting-LSTM-Normal	0.6058	0.6824	0.6444	0.7250	0.6423
GradientBoosting-LSTM-Absolute	0.5912	0.7383	0.5896	0.9875	0.5933
GradientBoosting-LSTM-Squared	0.5985	0.7442	0.5926	1.0000	0.5866
GradientBoosting-LSTM-Log	0.5912	0.7407	0.5882	1.0000	0.5421
GradientBoosting-BiLSTM-Normal	0.5620	0.6629	0.6020	0.7375	0.6147

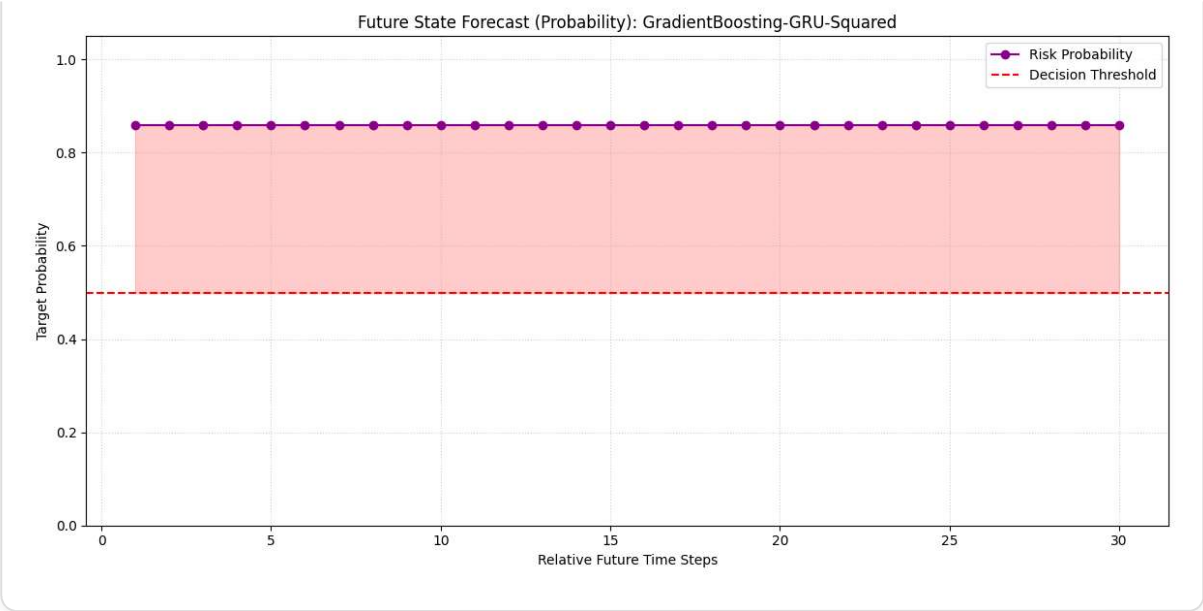
Model	Accuracy	F1	Precision	Recall	ROC-AUC
GradientBoosting-BiLSTM-Absolute	0.5912	0.7383	0.5896	0.9875	0.5329
GradientBoosting-BiLSTM-Squared	0.5839	0.7349	0.5852	0.9875	0.5966
GradientBoosting-BiLSTM-Log	0.5912	0.7407	0.5882	1.0000	0.5536
GradientBoosting-GRU-Normal	0.5985	0.6746	0.6404	0.7125	0.6239
GradientBoosting-GRU-Absolute	0.5839	0.7373	0.5839	1.0000	0.5606
GradientBoosting-GRU-Squared	0.6058	0.7453	0.5985	0.9875	0.6007
GradientBoosting-GRU-Log	0.5912	0.7407	0.5882	1.0000	0.5759
GradientBoosting-CNN-LSTM-Normal	0.5912	0.6706	0.6333	0.7125	0.6364
GradientBoosting-CNN-LSTM-Absolute	0.5985	0.7418	0.5940	0.9875	0.5875
GradientBoosting-CNN-LSTM-Squared	0.5912	0.7383	0.5896	0.9875	0.5904
GradientBoosting-CNN-LSTM-Log	0.5985	0.7442	0.5926	1.0000	0.5695
SVM-LSTM-Normal	0.5985	0.6821	0.6344	0.7375	0.6123
SVM-LSTM-Absolute	0.5839	0.7373	0.5839	1.0000	0.4594
SVM-LSTM-Squared	0.5839	0.7373	0.5839	1.0000	0.4250

Model	Accuracy	F1	Precision	Recall	ROC-AUC
SVM-LSTM-Log	0.5839	0.7373	0.5839	1.0000	0.4553
SVM-BiLSTM-Normal	0.5839	0.6415	0.6456	0.6375	0.6445
SVM-BiLSTM-Absolute	0.5839	0.7373	0.5839	1.0000	0.4873
SVM-BiLSTM-Squared	0.5839	0.7373	0.5839	1.0000	0.4292
SVM-BiLSTM-Log	0.5839	0.7373	0.5839	1.0000	0.4978
SVM-GRU-Normal	0.5985	0.6961	0.6238	0.7875	0.6140
SVM-GRU-Absolute	0.5839	0.7373	0.5839	1.0000	0.5700
SVM-GRU-Squared	0.5839	0.7373	0.5839	1.0000	0.5197
SVM-GRU-Log	0.5839	0.7373	0.5839	1.0000	0.4799
SVM-CNN-LSTM-Normal	0.6058	0.6824	0.6444	0.7250	0.6547
SVM-CNN-LSTM-Absolute	0.5839	0.7373	0.5839	1.0000	0.5081
SVM-CNN-LSTM-Squared	0.5839	0.7373	0.5839	1.0000	0.5004
SVM-CNN-LSTM-Log	0.5839	0.7373	0.5839	1.0000	0.4743
MLP-LSTM-Normal	0.5985	0.6667	0.6471	0.6875	0.6452
MLP-LSTM-Absolute	0.6058	0.7327	0.6066	0.9250	0.5717
MLP-LSTM-Squared	0.5766	0.7238	0.5846	0.9500	0.5931
MLP-LSTM-Log	0.5766	0.7264	0.5833	0.9625	0.6024
MLP-BiLSTM-Normal	0.5985	0.6707	0.6437	0.7000	0.6618

Model	Accuracy	F1	Precision	Recall	ROC-AUC
MLP-BiLSTM-Absolute	0.5985	0.7317	0.6000	0.9375	0.5614
MLP-BiLSTM-Squared	0.5839	0.7246	0.5906	0.9375	0.5940
MLP-BiLSTM-Log	0.5839	0.7246	0.5906	0.9375	0.6061
MLP-GRU-Normal	0.5912	0.6585	0.6429	0.6750	0.6509
MLP-GRU-Absolute	0.5839	0.7192	0.5935	0.9125	0.5738
MLP-GRU-Squared	0.5839	0.7324	0.5865	0.9750	0.5976
MLP-GRU-Log	0.5839	0.7220	0.5920	0.9250	0.6060
MLP-CNN-LSTM-Normal	0.6204	0.6867	0.6628	0.7125	0.6430
MLP-CNN-LSTM-Absolute	0.5912	0.7255	0.5968	0.9250	0.5800
MLP-CNN-LSTM-Squared	0.5912	0.7358	0.5909	0.9750	0.5942
MLP-CNN-LSTM-Log	0.5839	0.7246	0.5906	0.9375	0.5939
Ensemble	0.5912	0.7383	0.5896	0.9875	0.6384

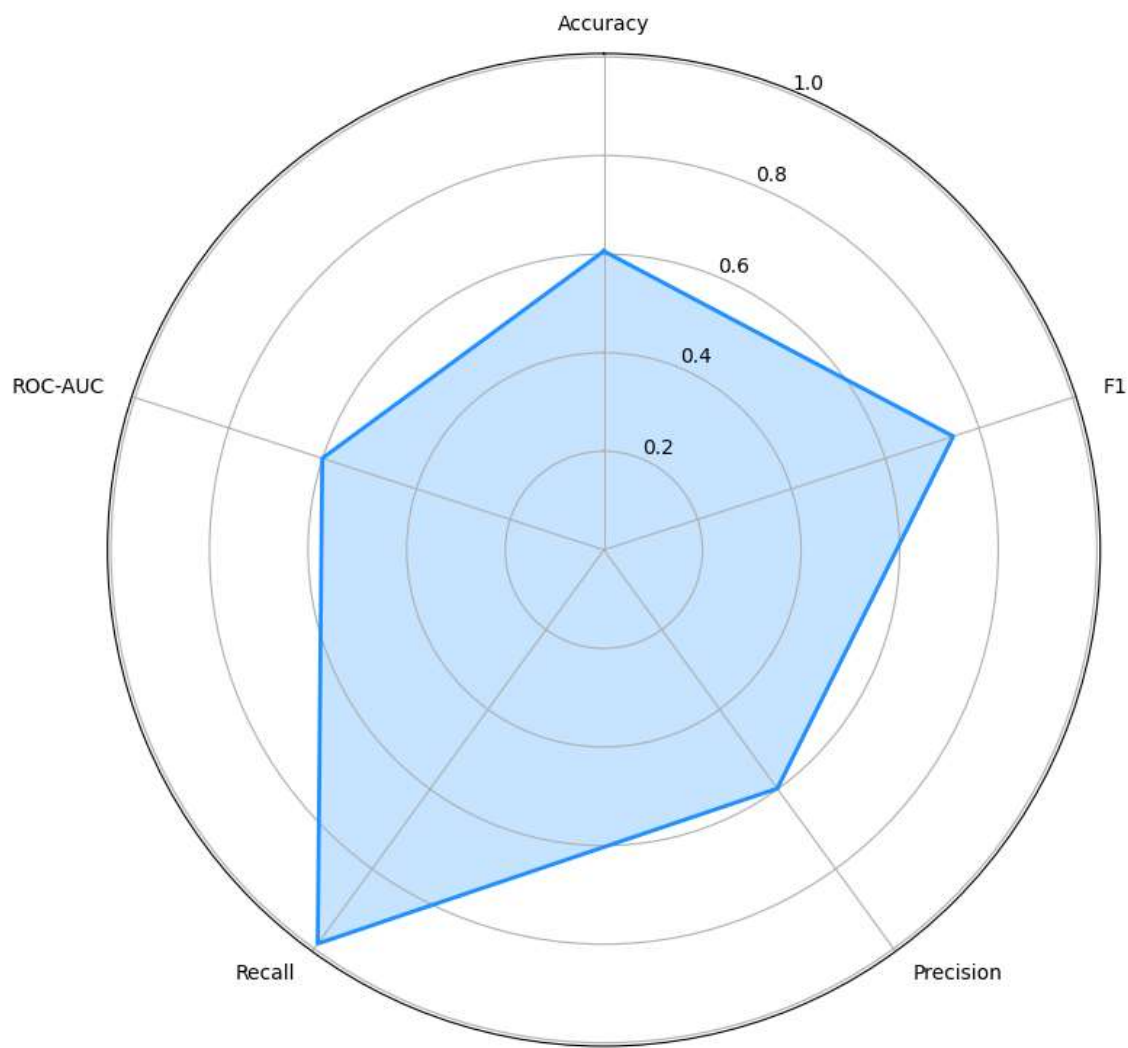
3. Strategic Insights: GradientBoosting-GRU-Squared (Best Performer)

Future Risk Probability Forecast



Radar: Performance Balance Profile

Performance Balance Profile: GradientBoosting-GRU-Squared

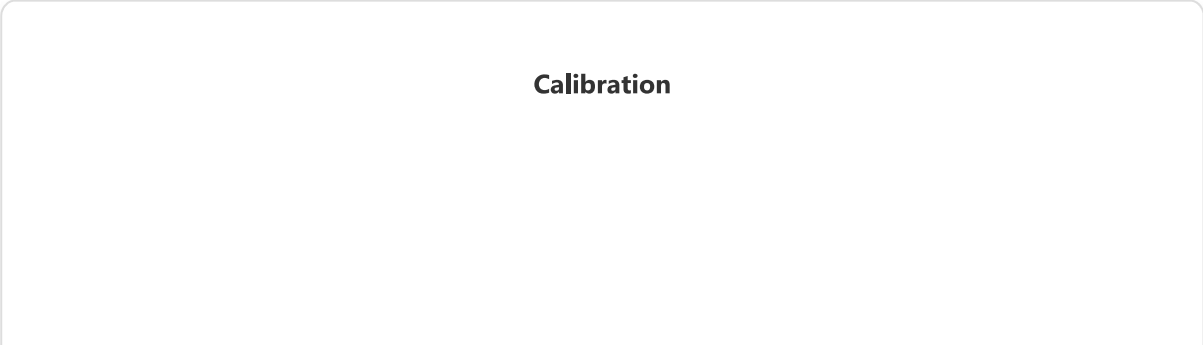
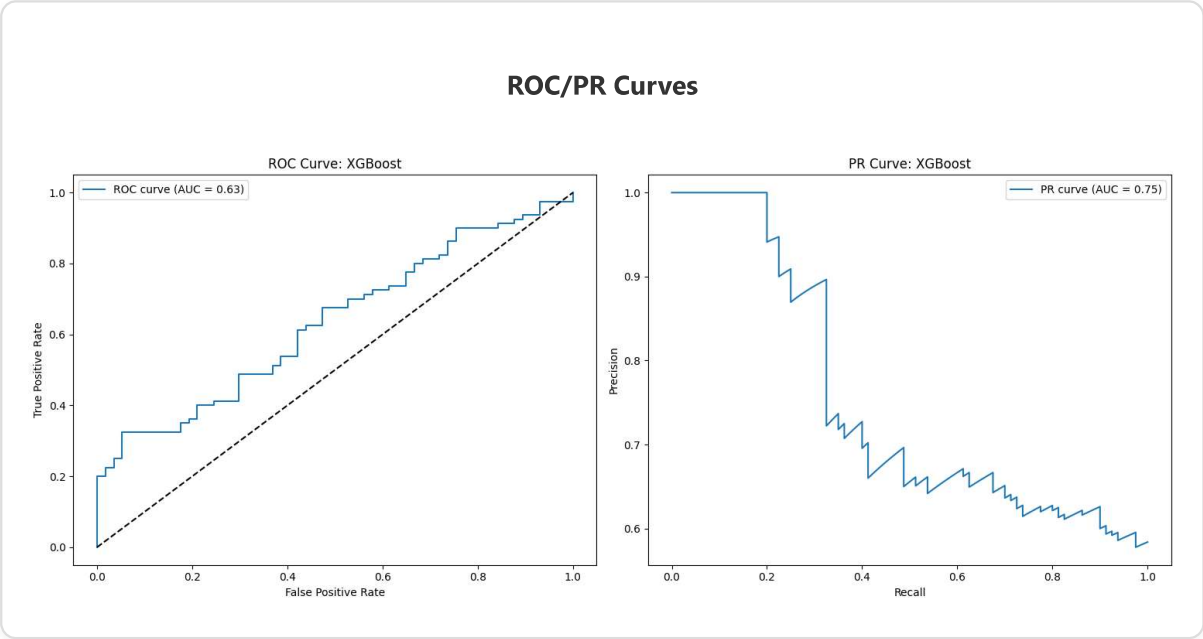
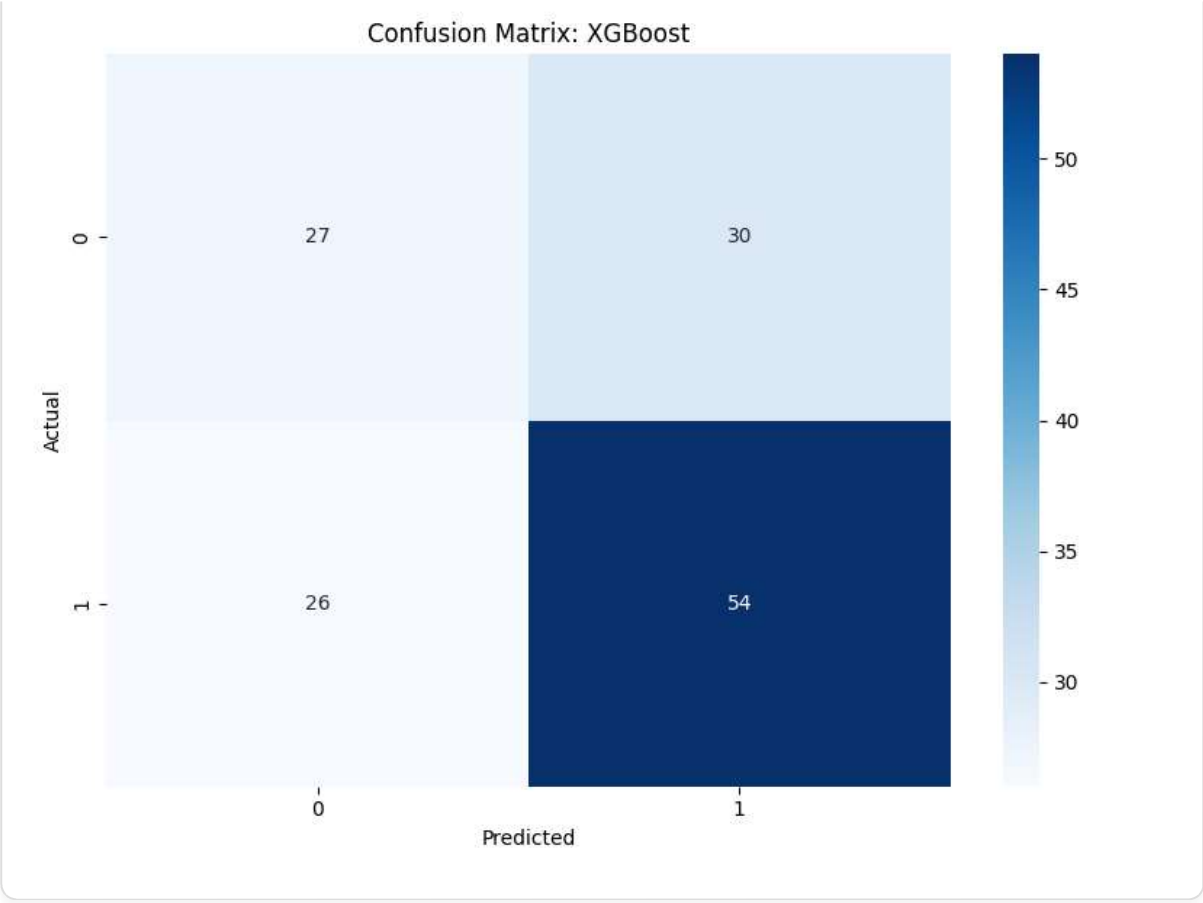


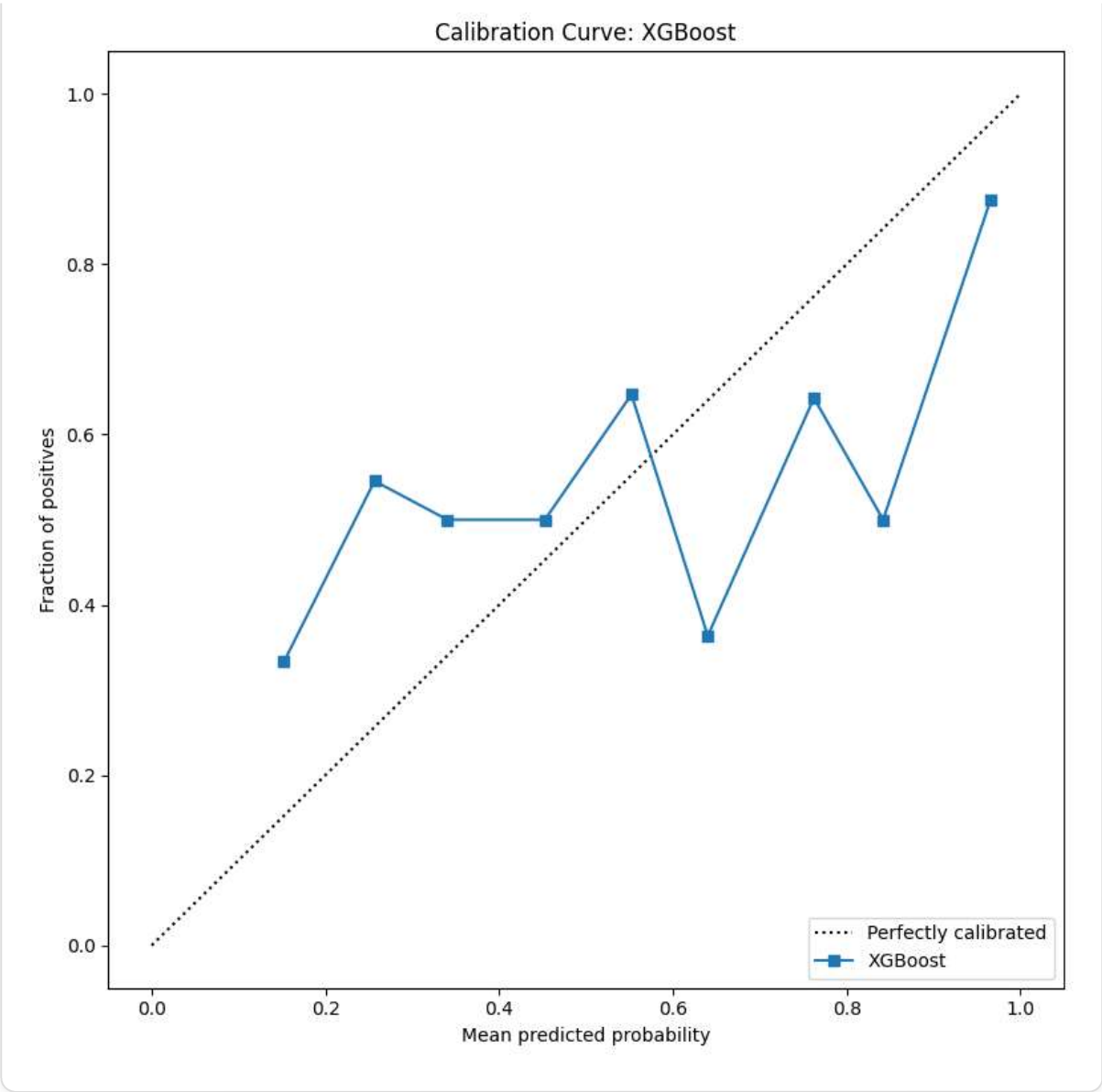
4. Detailed Model Diagnostics

Model: XGBoost

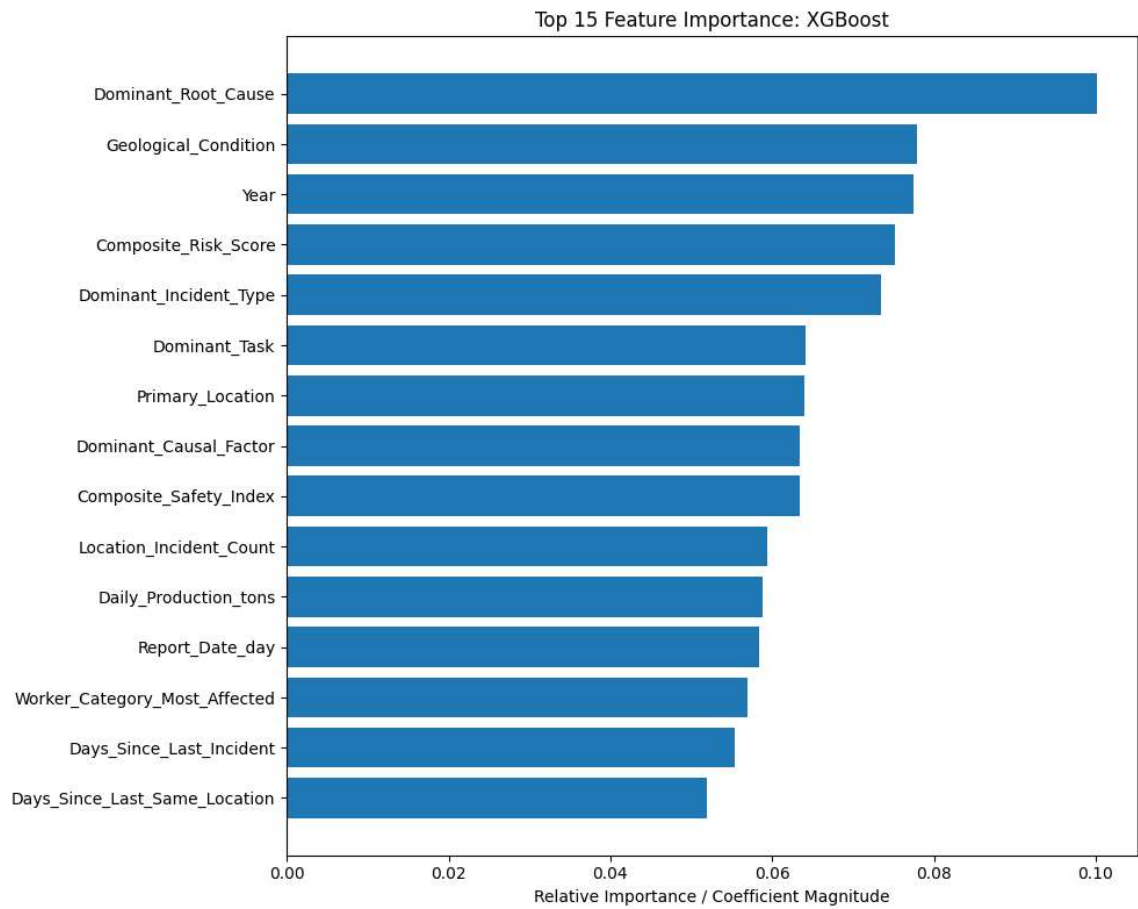
Performance Summary: Acc: 0.5912 | F1: 0.6585 | Prec: 0.6429 | Rec: 0.6750 | AUC: 0.6329 | MSE: 0.255171

Confusion Matrix



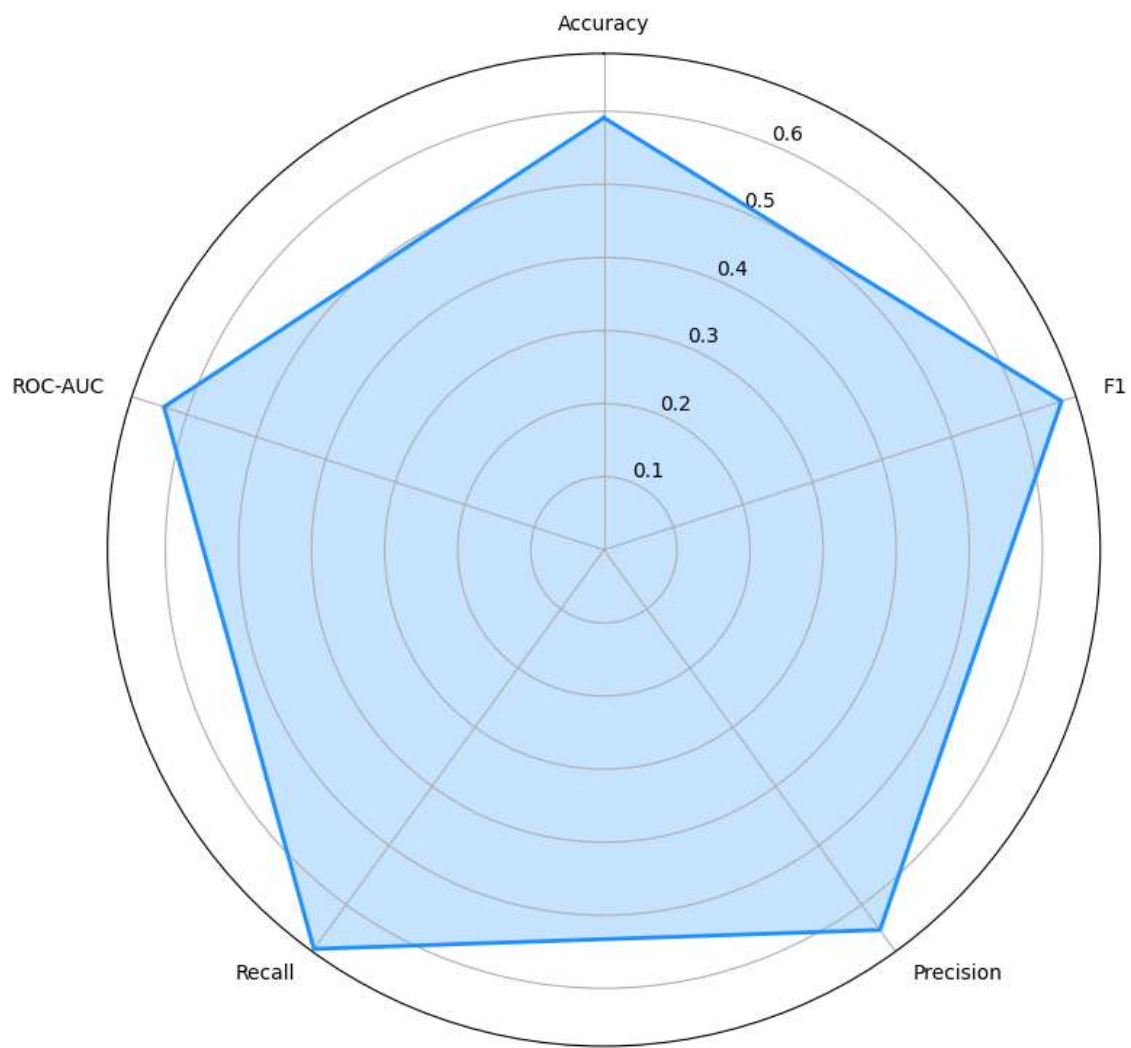


Feature Importance



Balance Radar

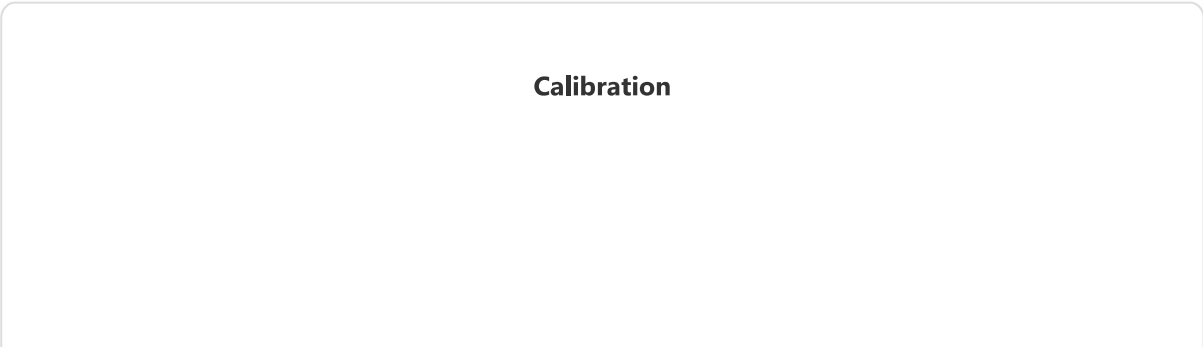
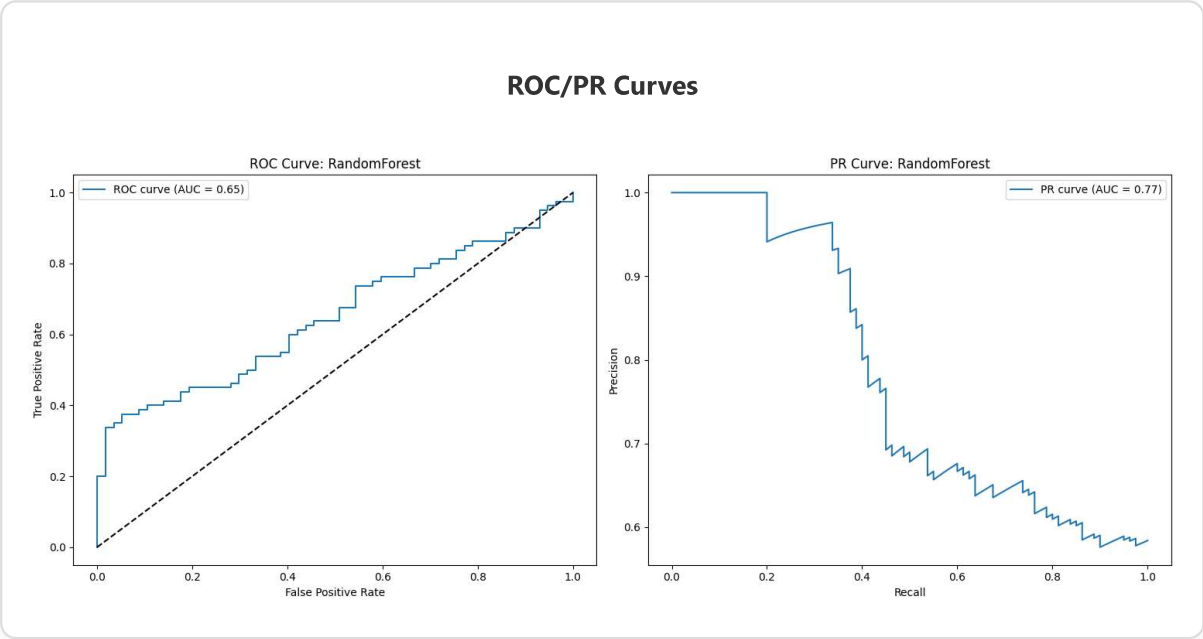
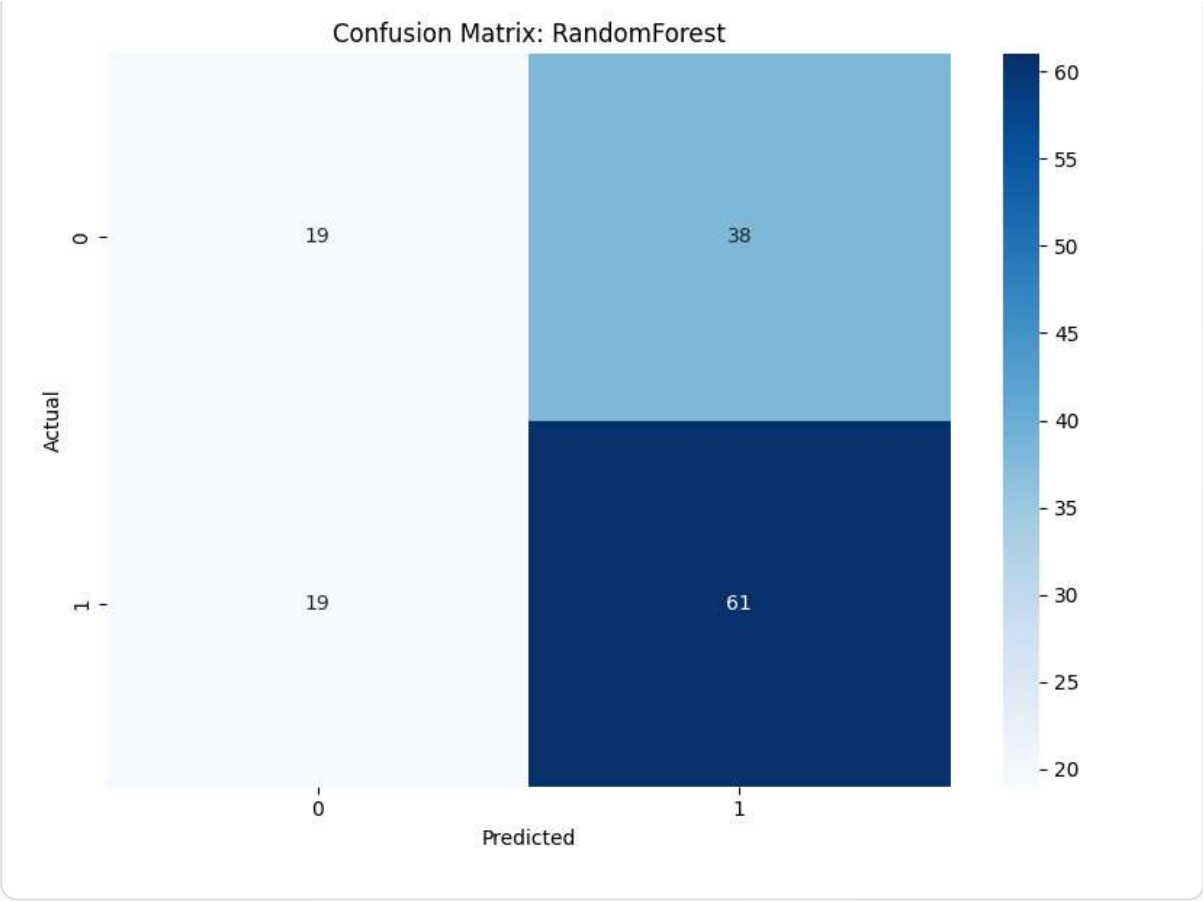
Performance Balance Profile: XGBoost

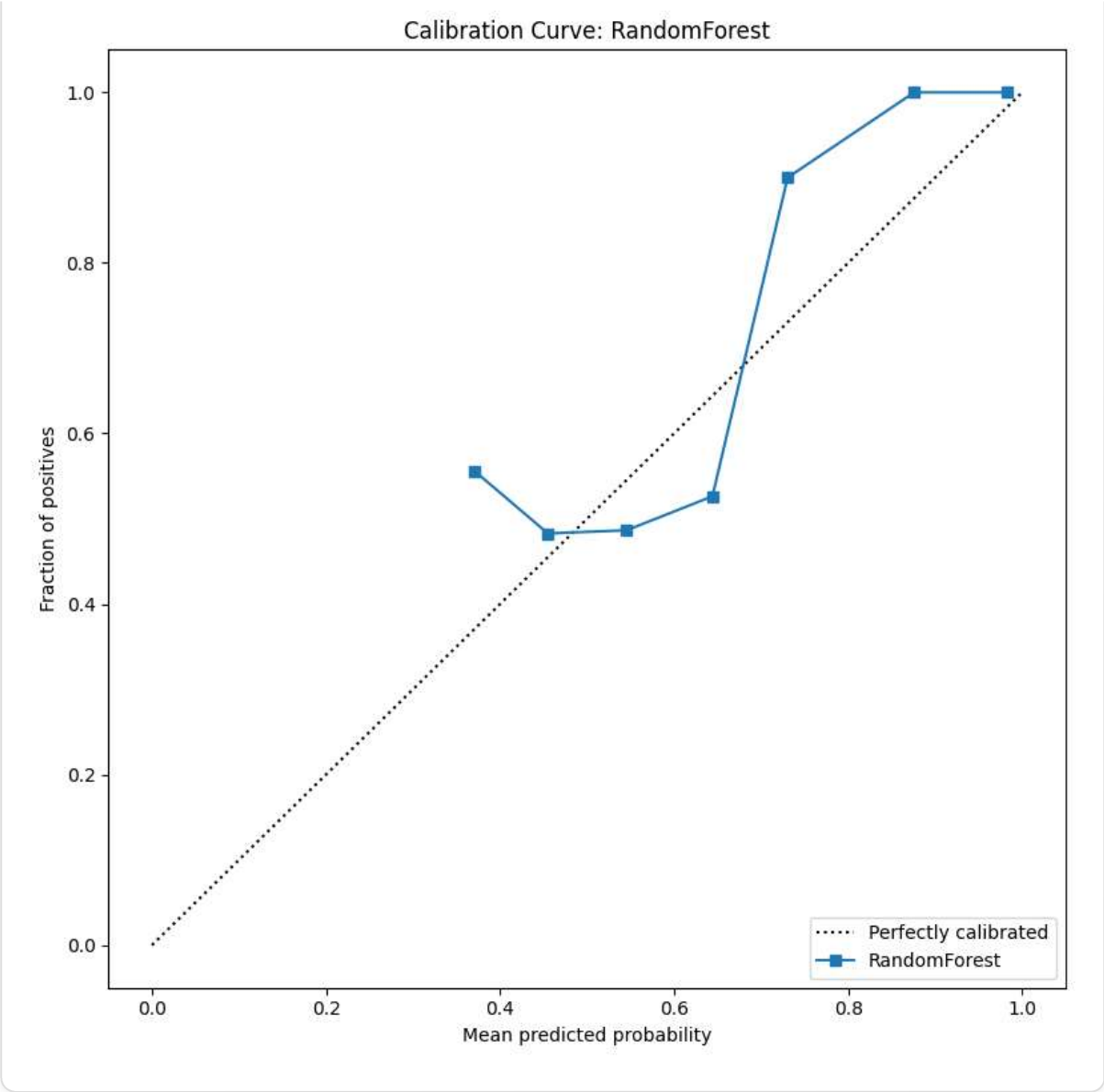


Model: RandomForest

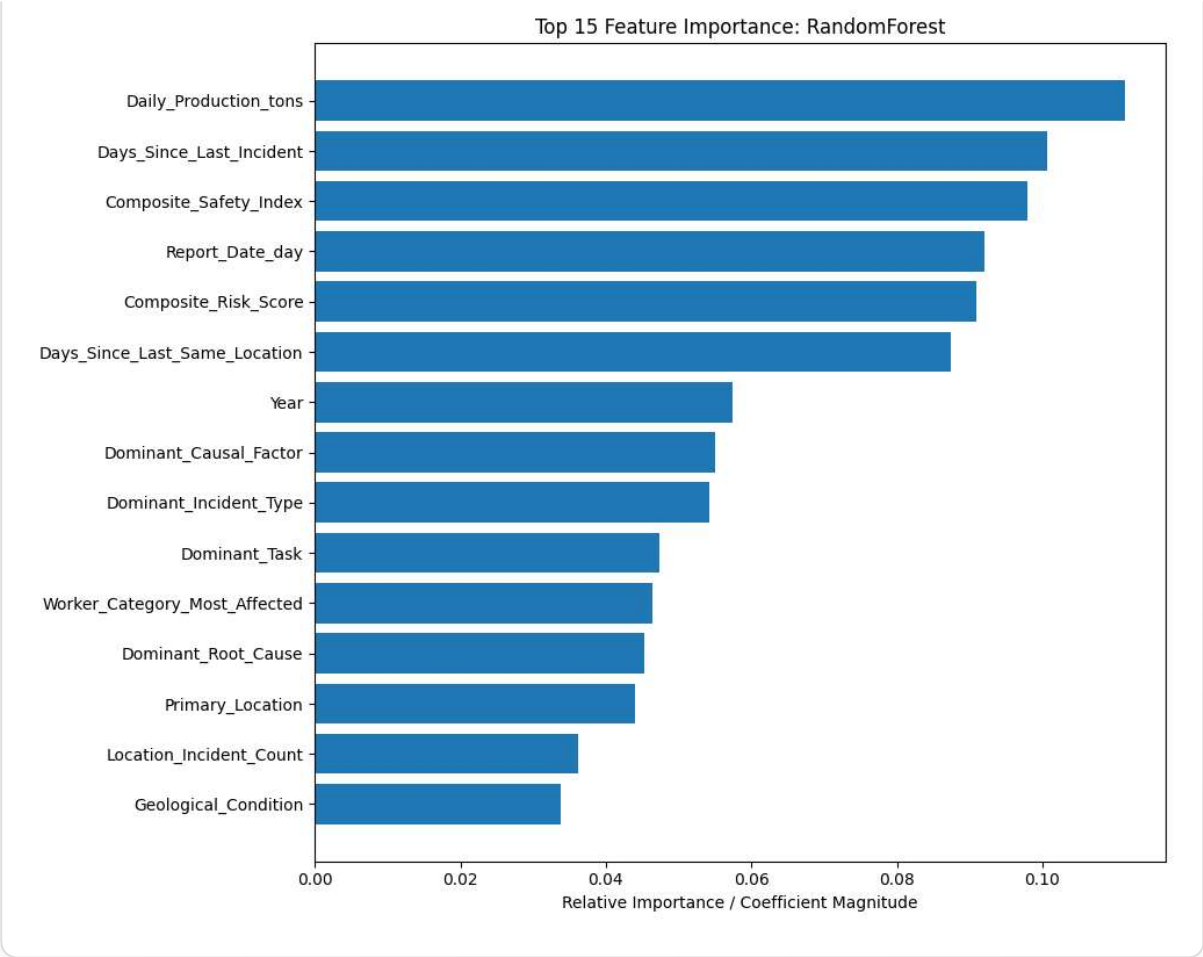
Performance Summary: Acc: 0.5839 | F1: 0.6816 | Prec: 0.6162 | Rec: 0.7625 | AUC: 0.6476 | MSE: 0.221620

Confusion Matrix



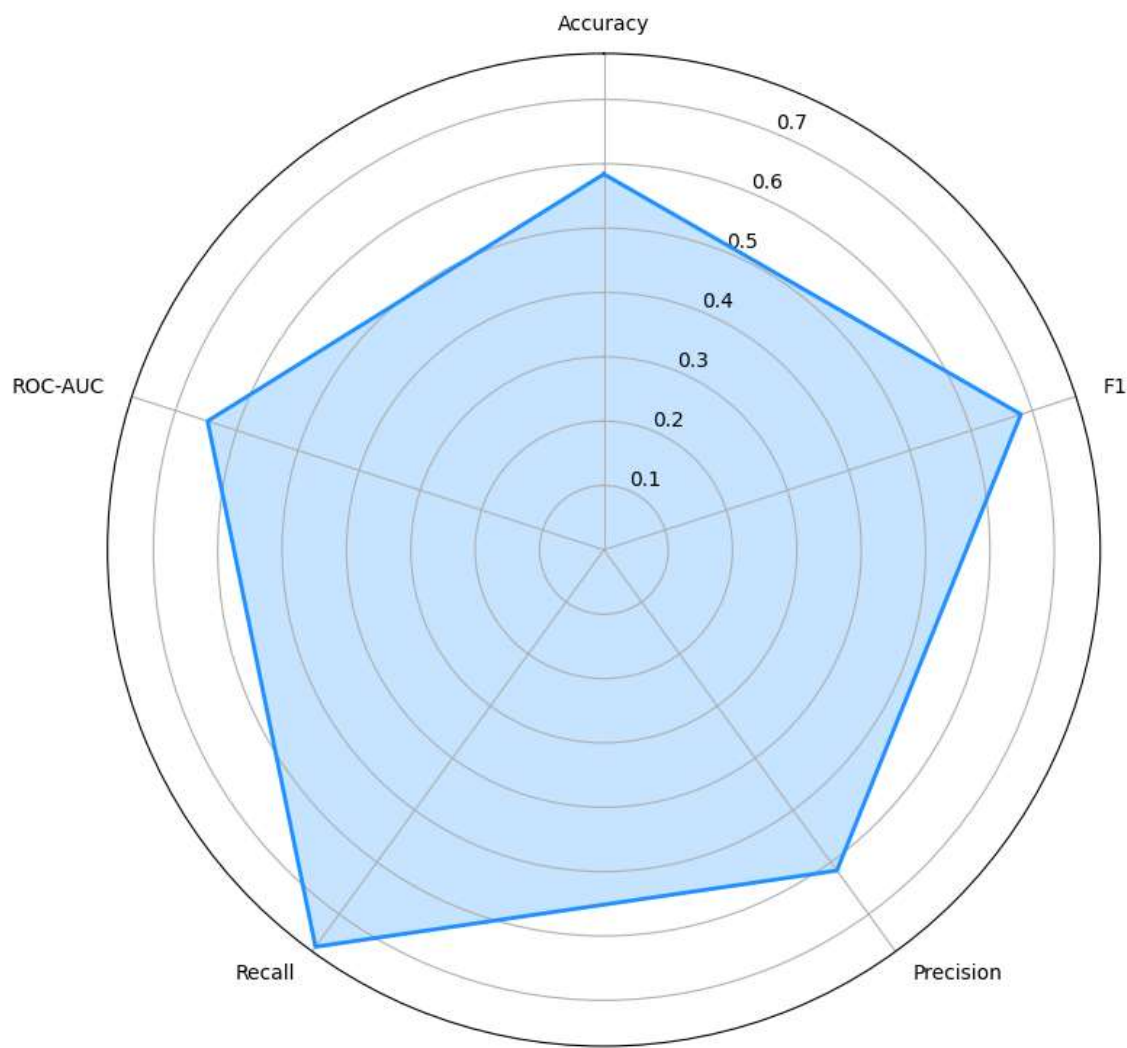


Feature Importance



Balance Radar

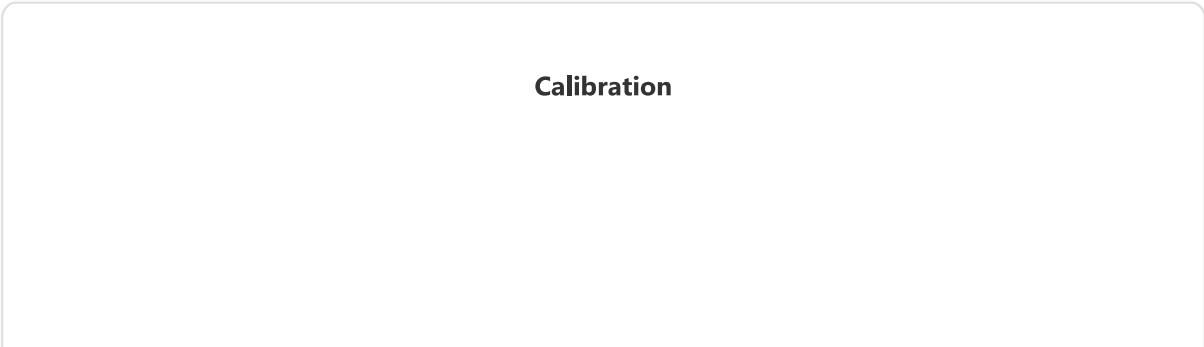
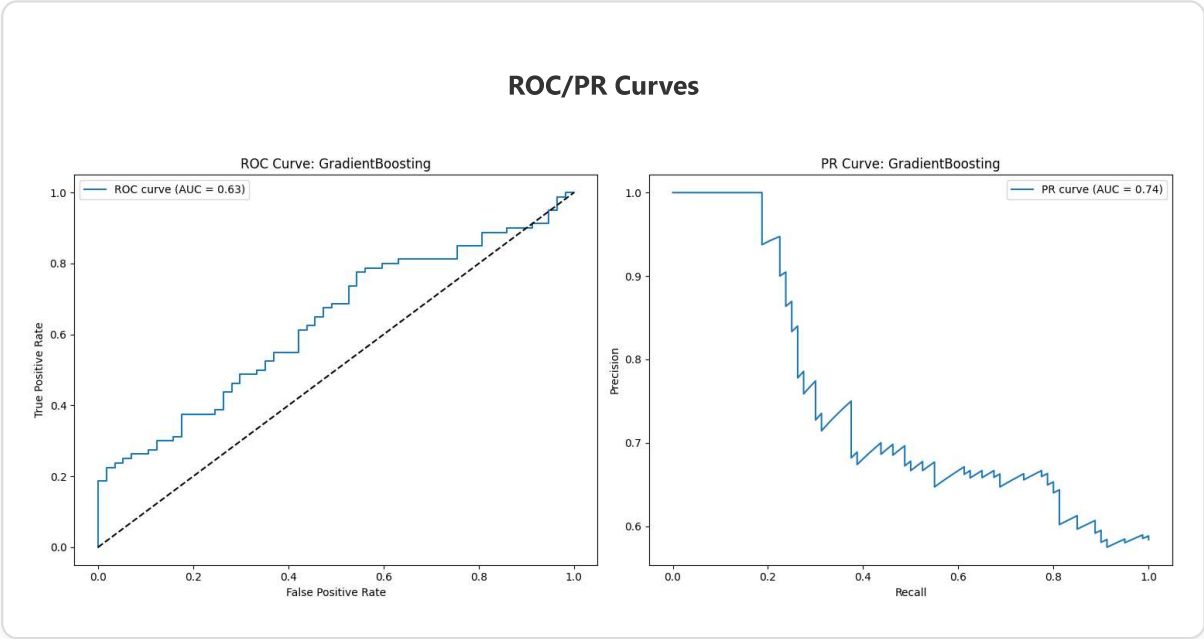
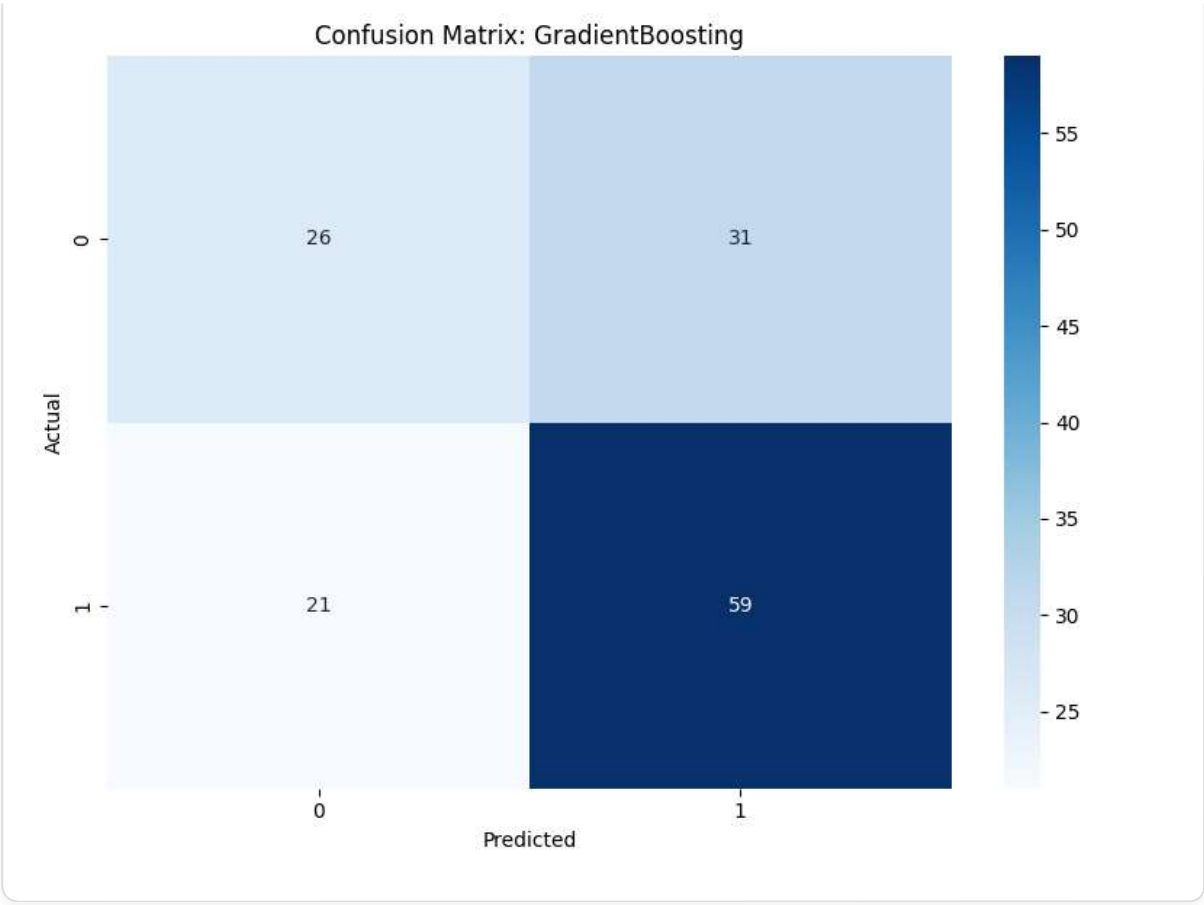
Performance Balance Profile: RandomForest

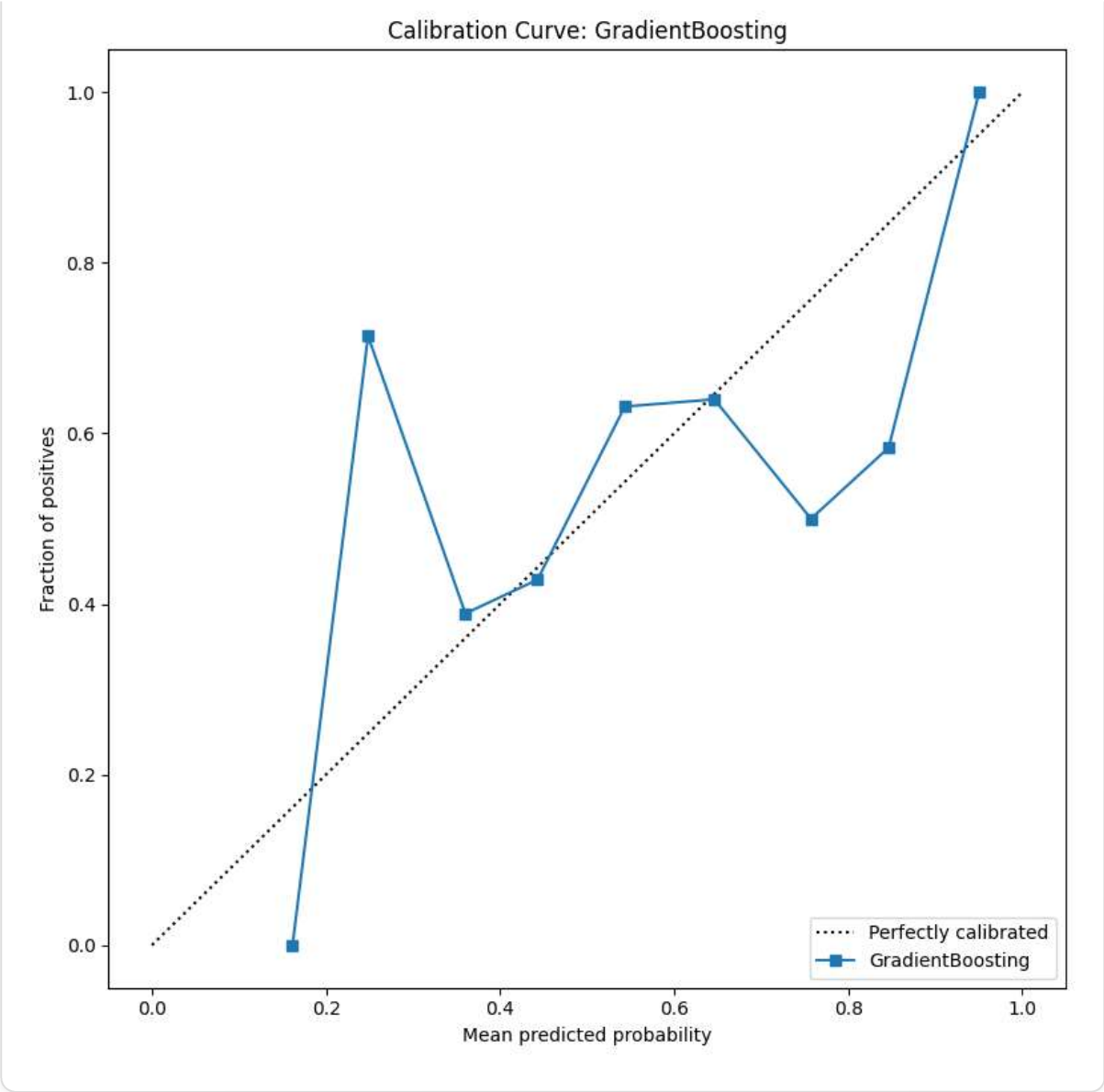


Model: GradientBoosting

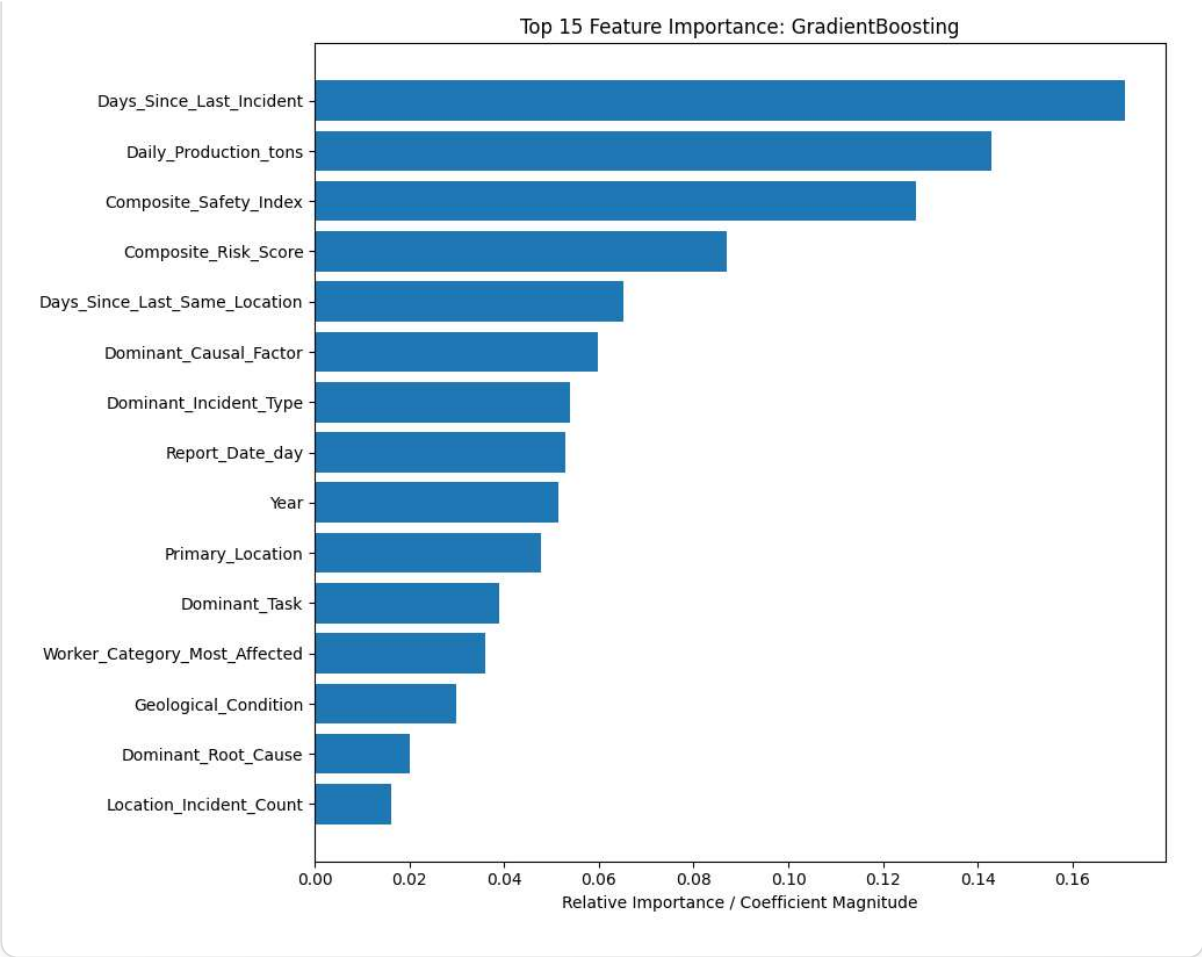
Performance Summary: Acc: 0.6204 | F1: 0.6941 | Prec: 0.6556 | Rec: 0.7375 | AUC: 0.6322 | MSE: 0.237347

Confusion Matrix



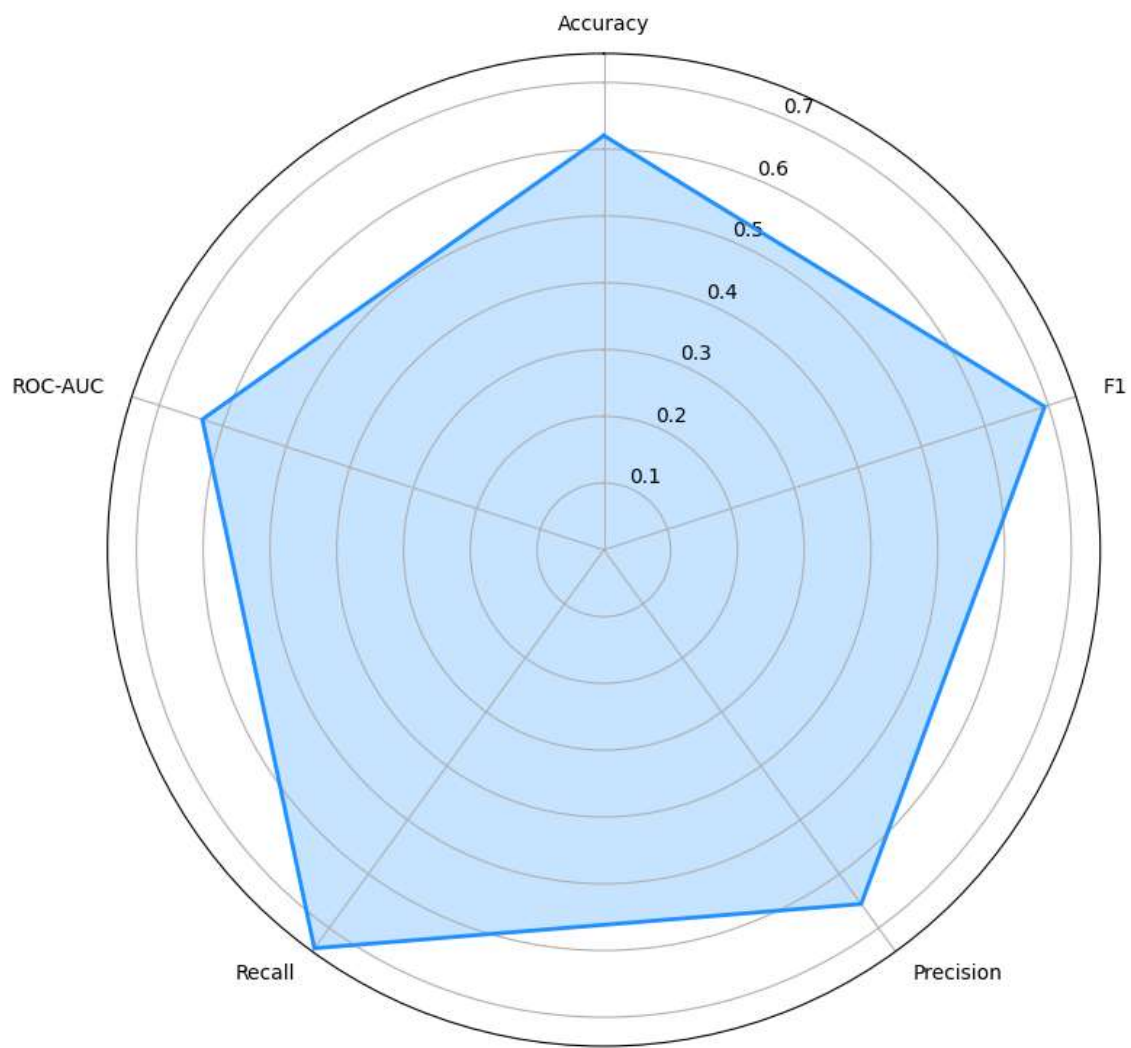


Feature Importance



Balance Radar

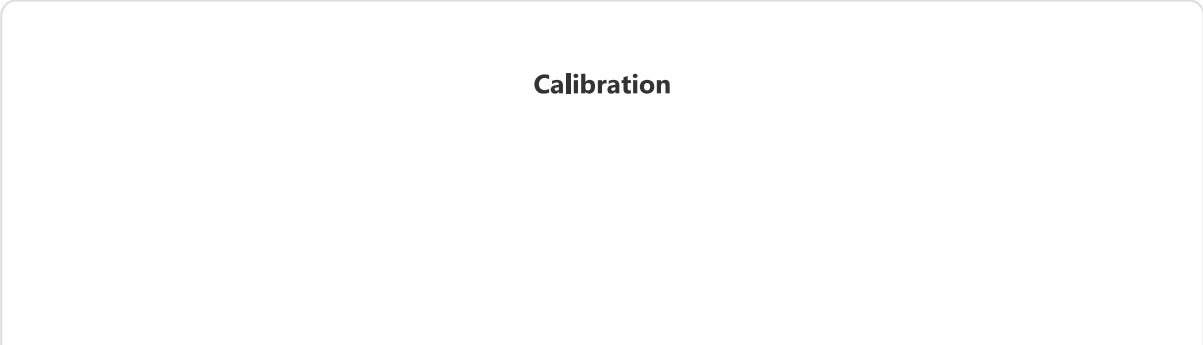
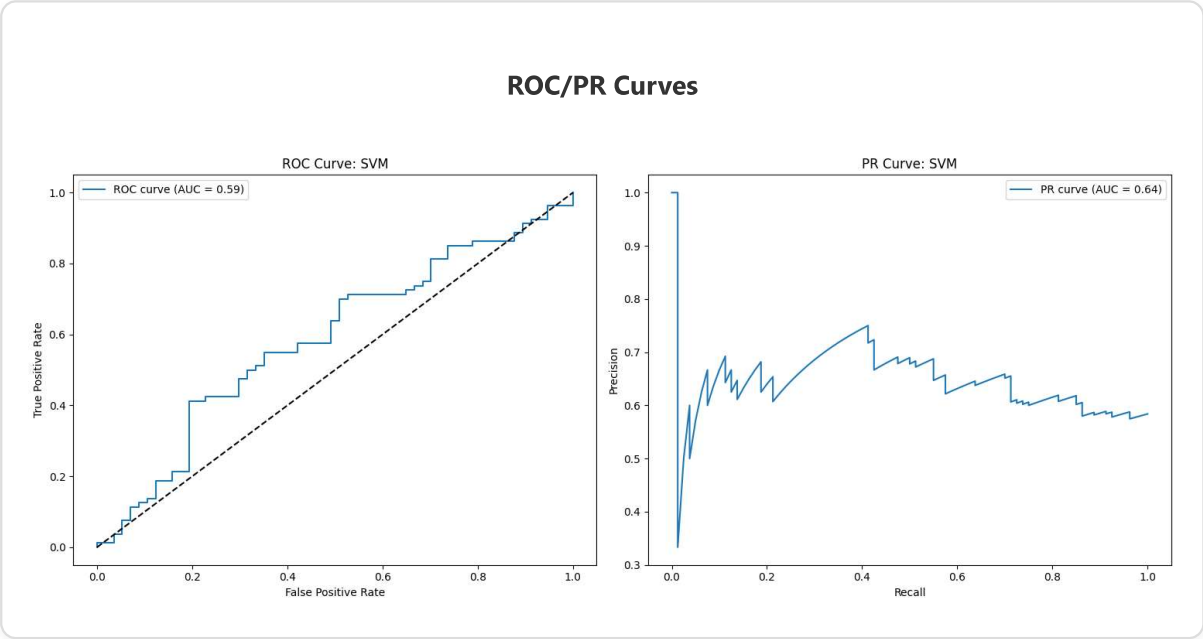
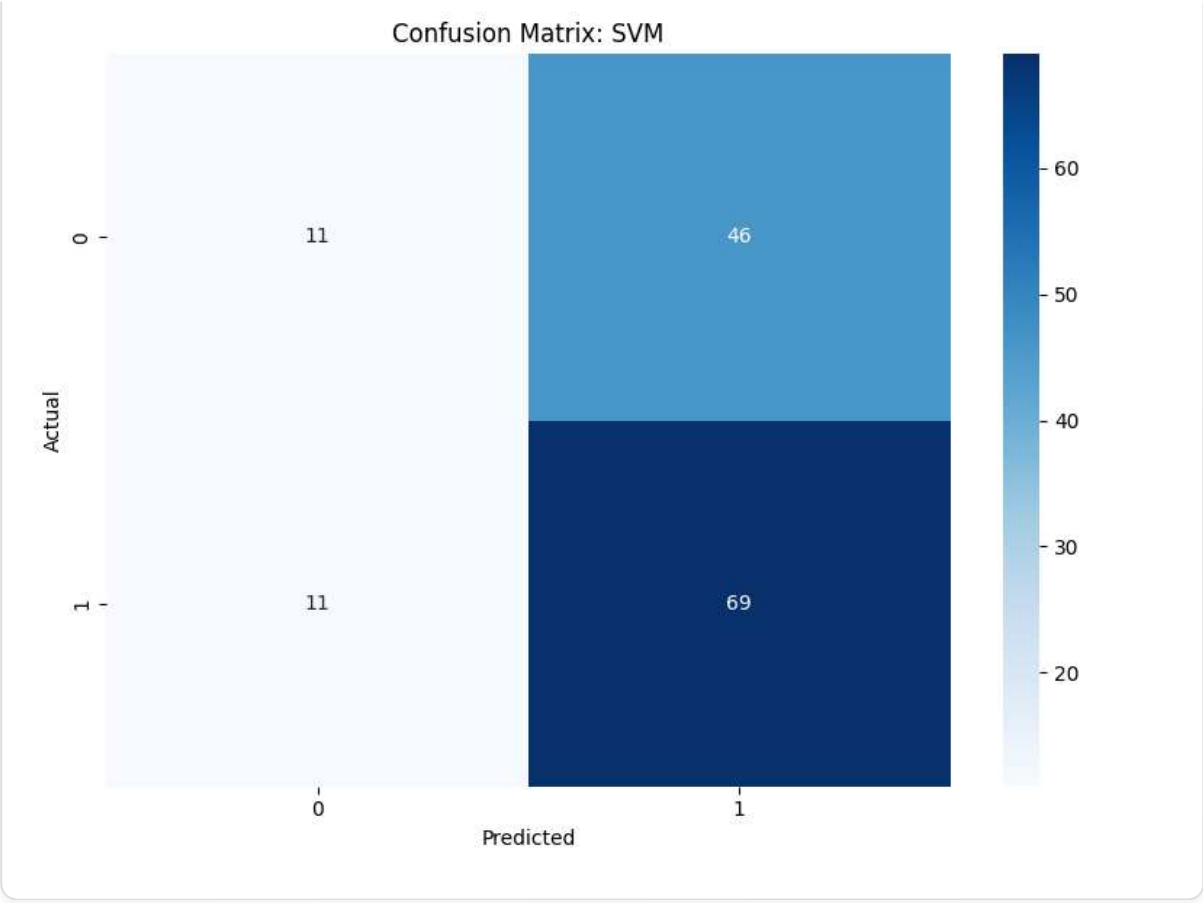
Performance Balance Profile: GradientBoosting

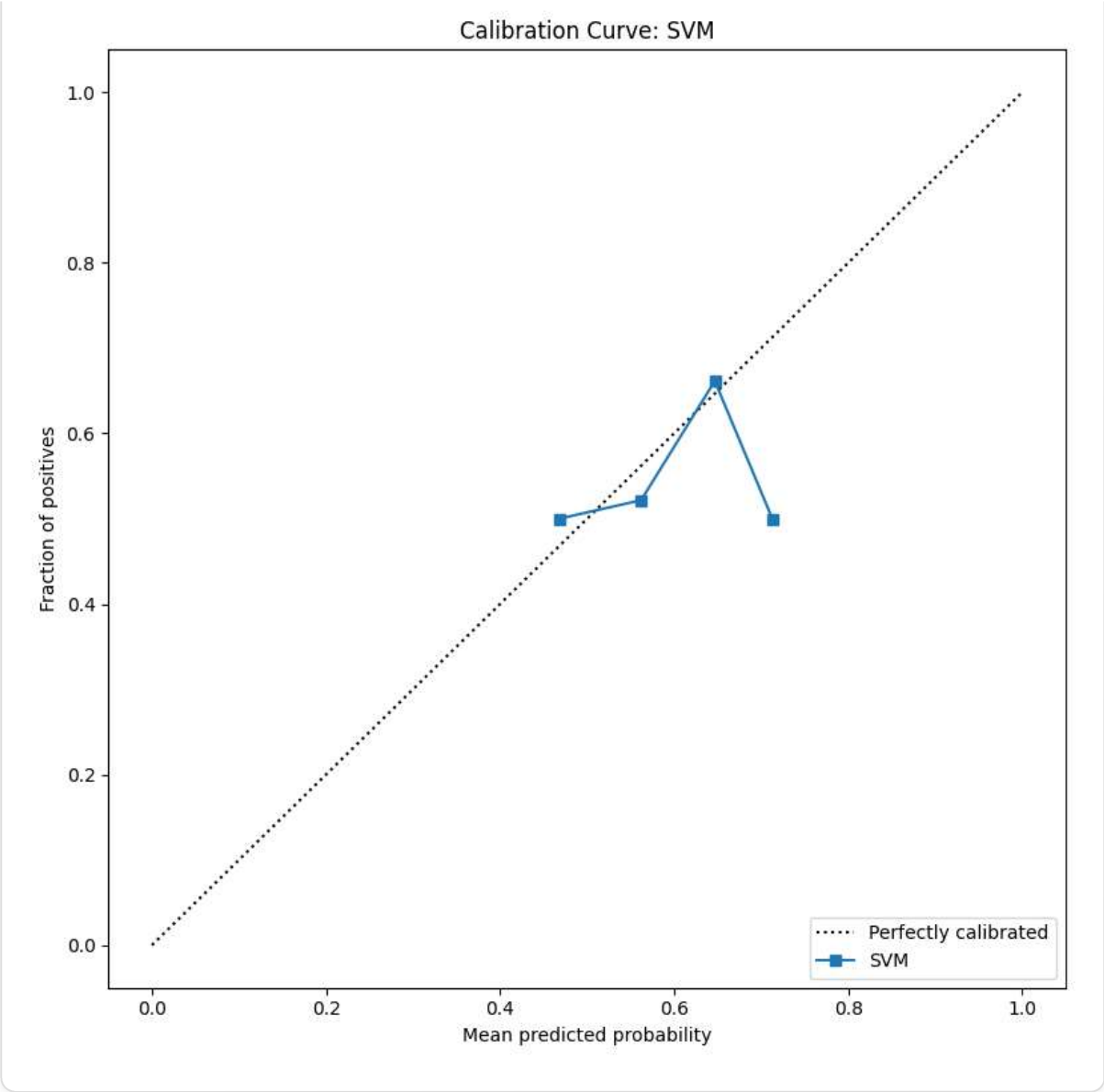


Model: SVM

Performance Summary: Acc: 0.5839 | F1: 0.7077 | Prec: 0.6000 | Rec: 0.8625 | AUC: 0.5857 | MSE: 0.238969

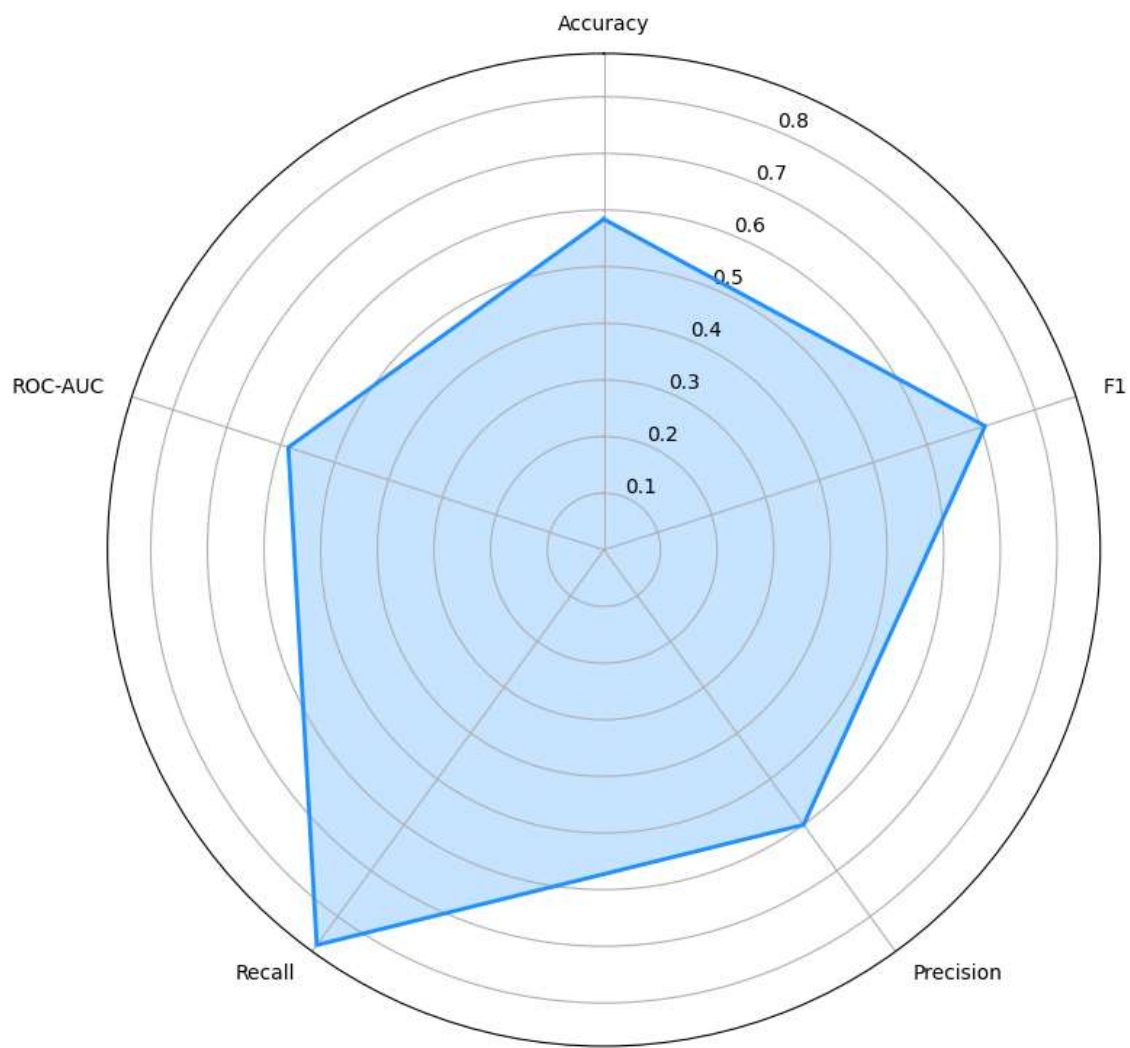
Confusion Matrix





Balance Radar

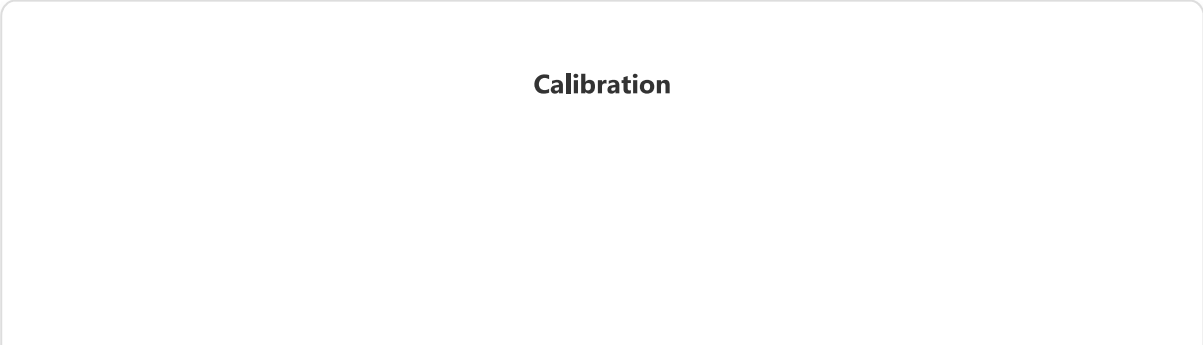
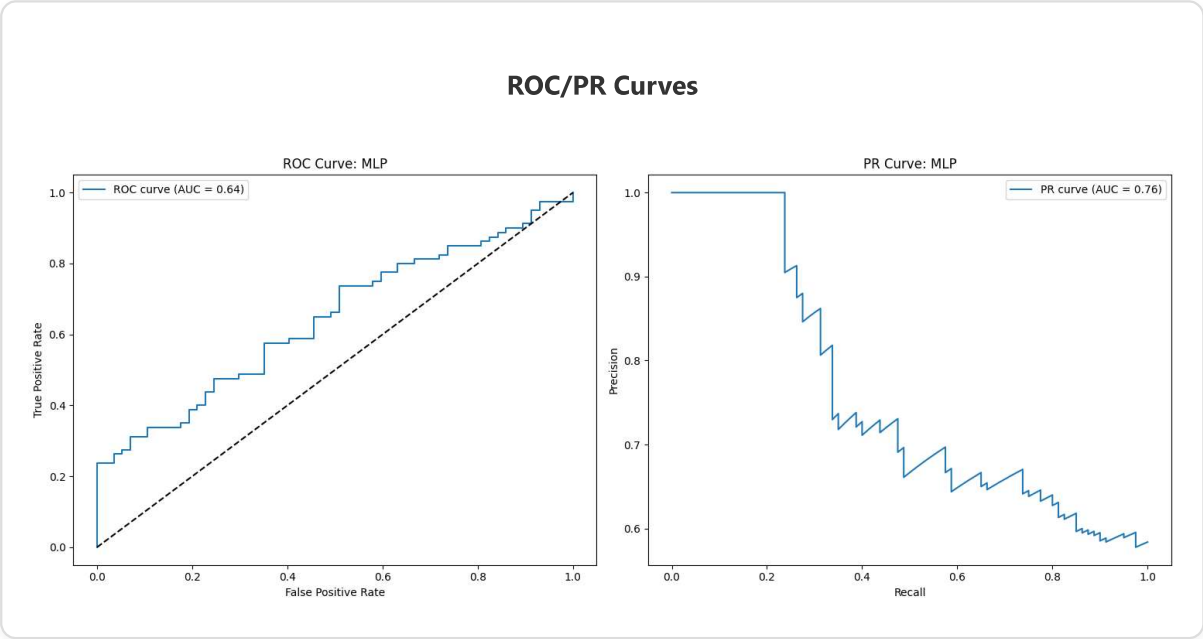
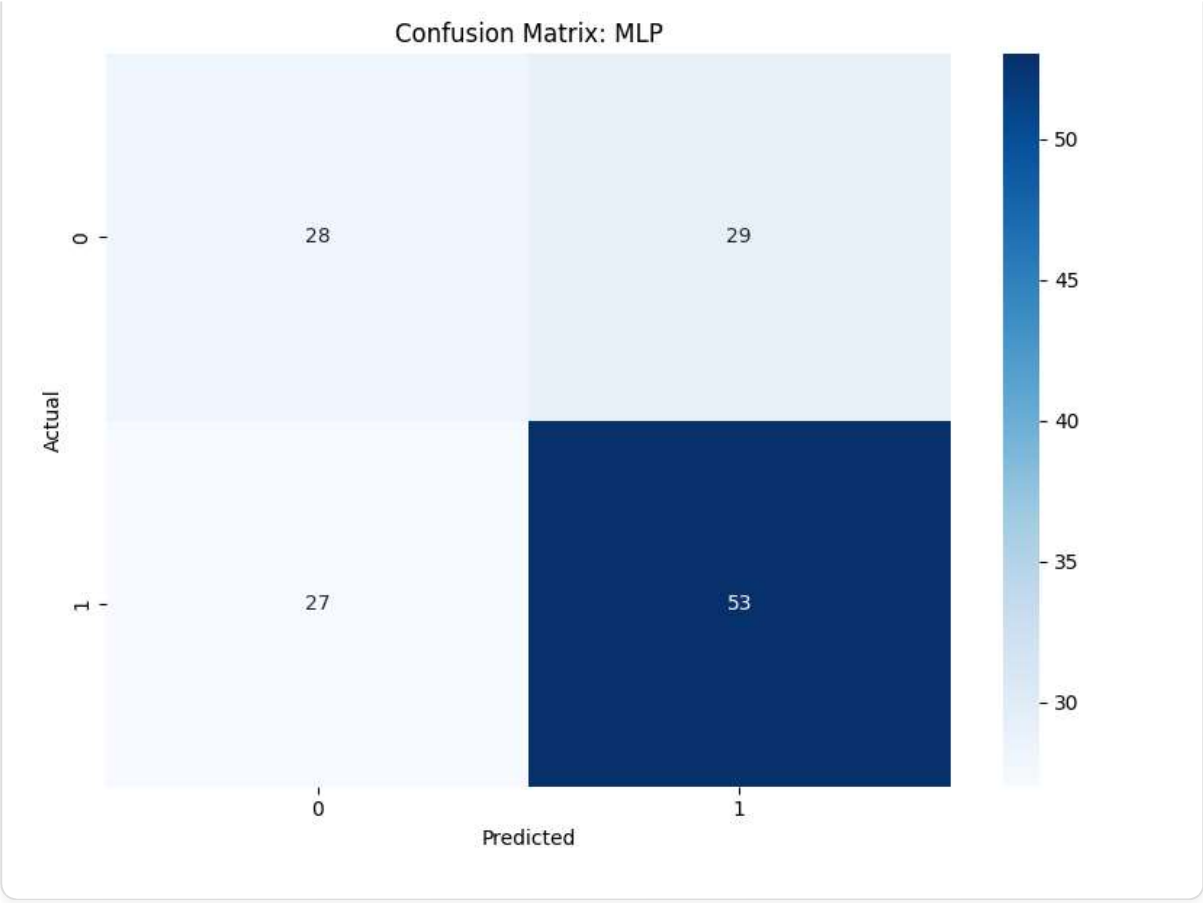
Performance Balance Profile: SVM

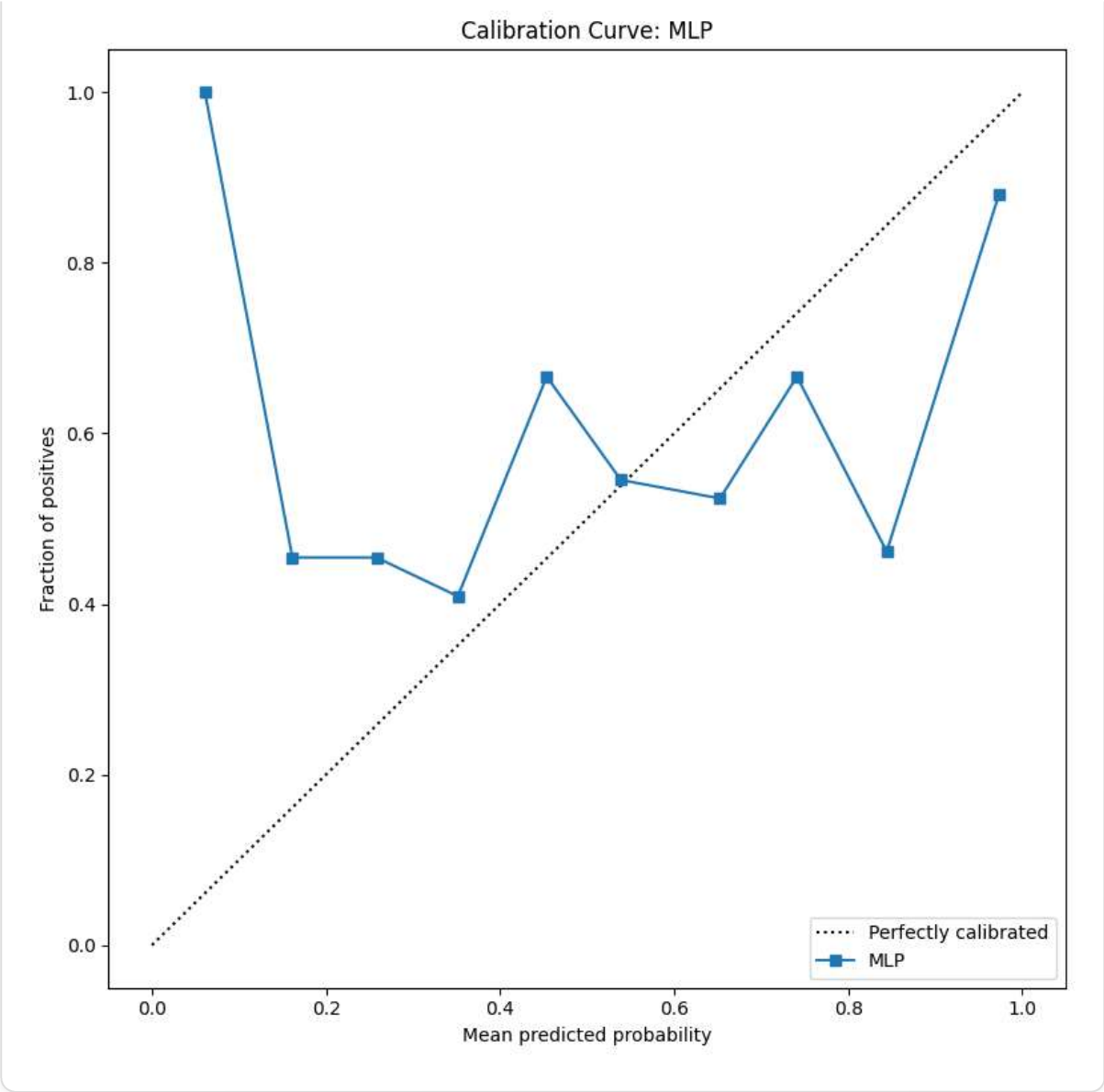


Model: MLP

Performance Summary: Acc: 0.5912 | F1: 0.6543 | Prec: 0.6463 | Rec: 0.6625 | AUC: 0.6421 | MSE: 0.253480

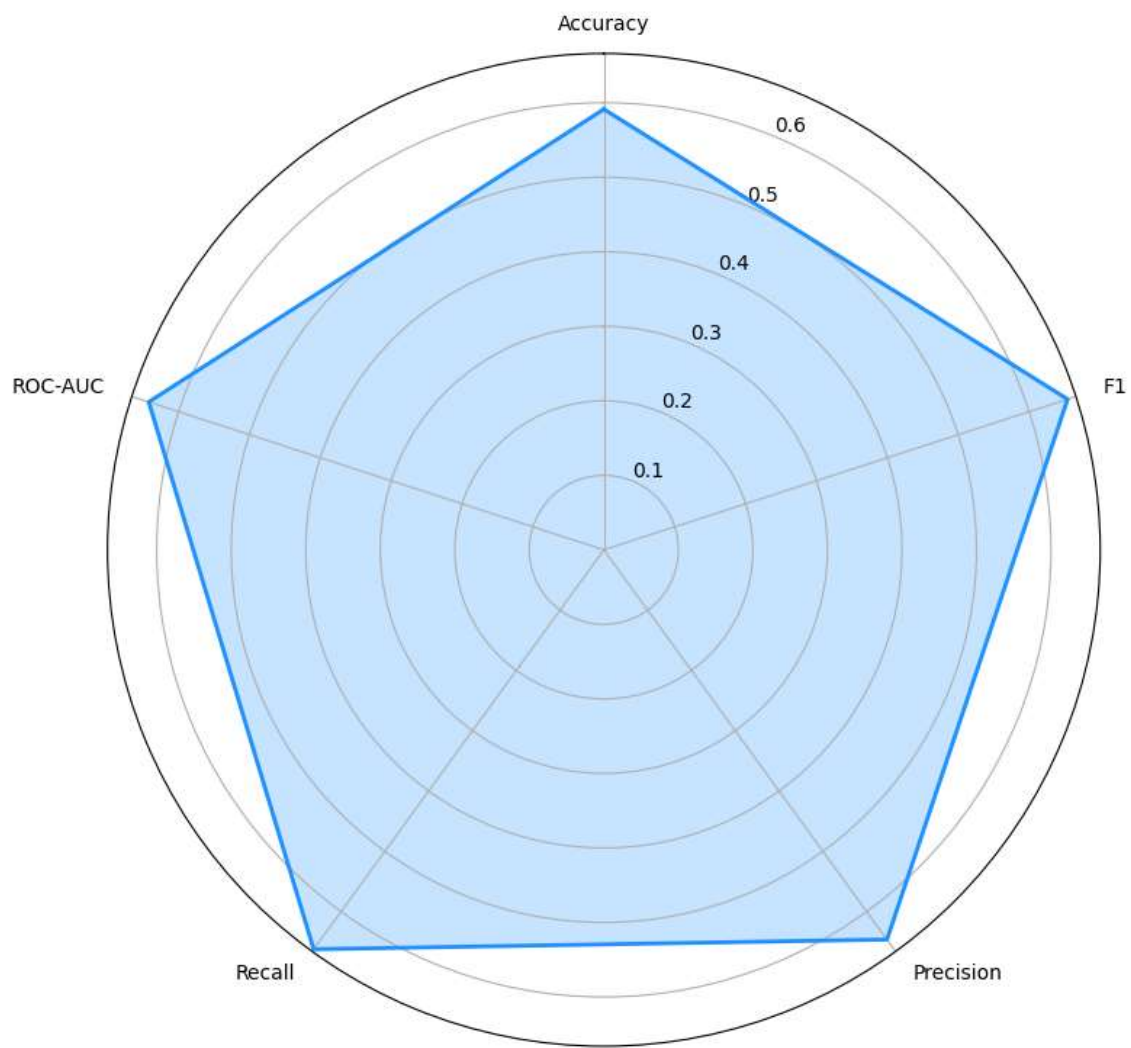
Confusion Matrix





Balance Radar

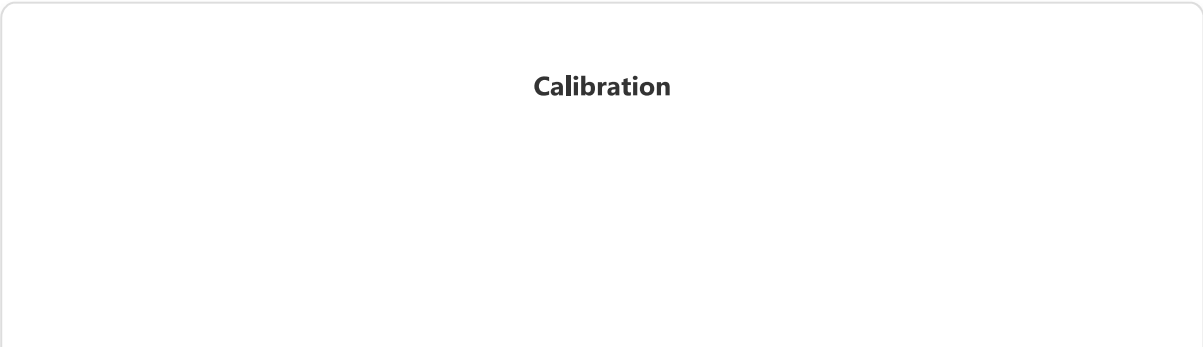
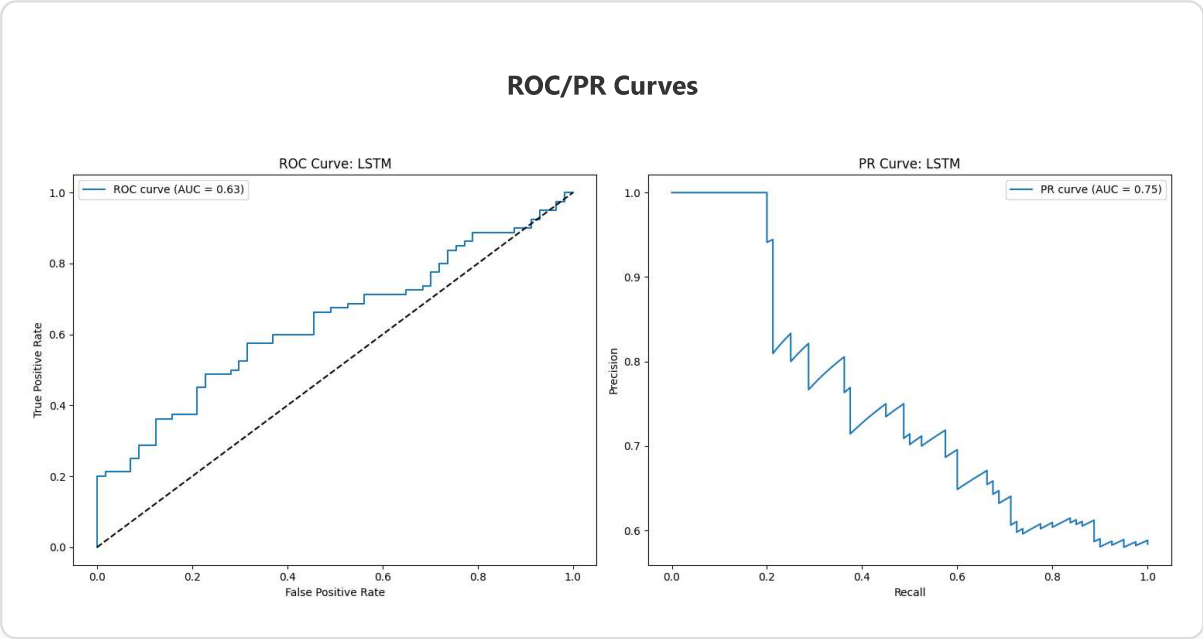
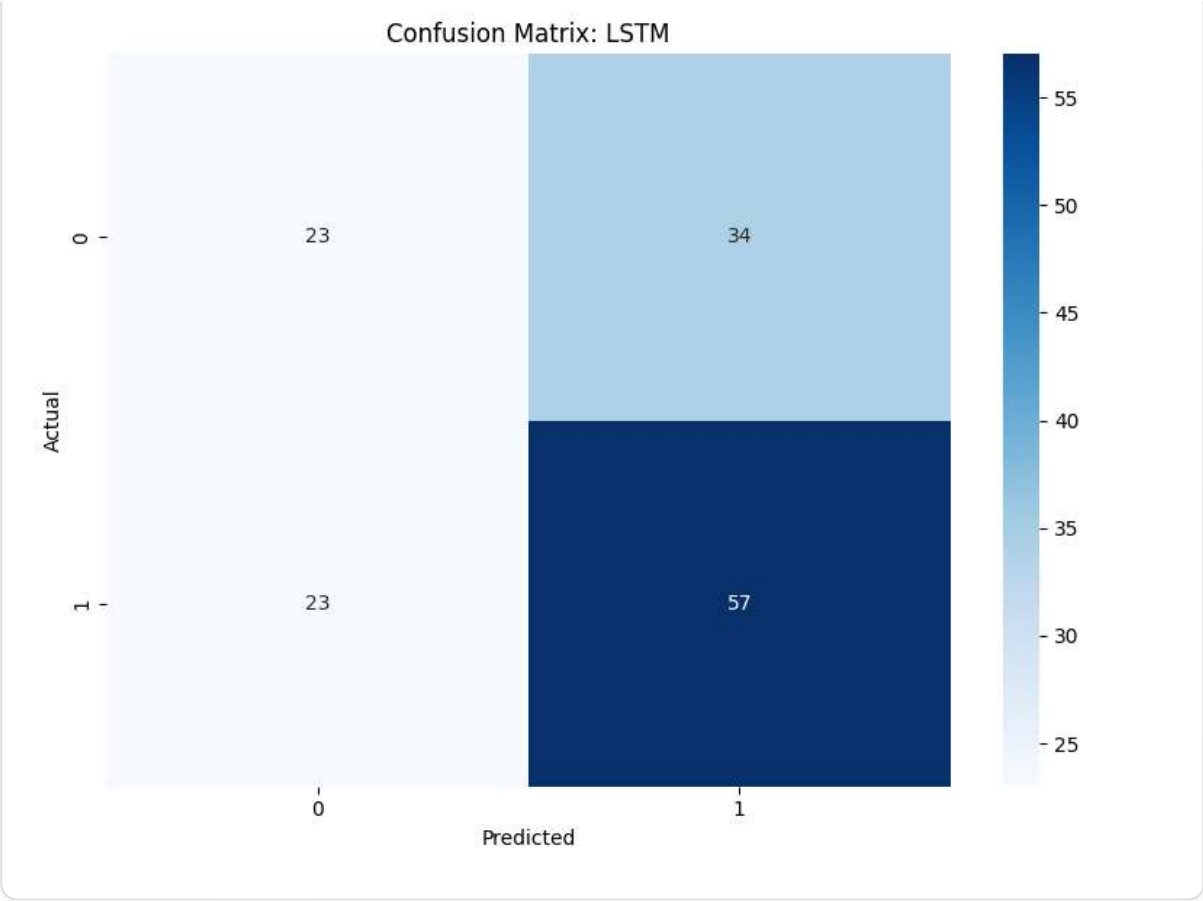
Performance Balance Profile: MLP

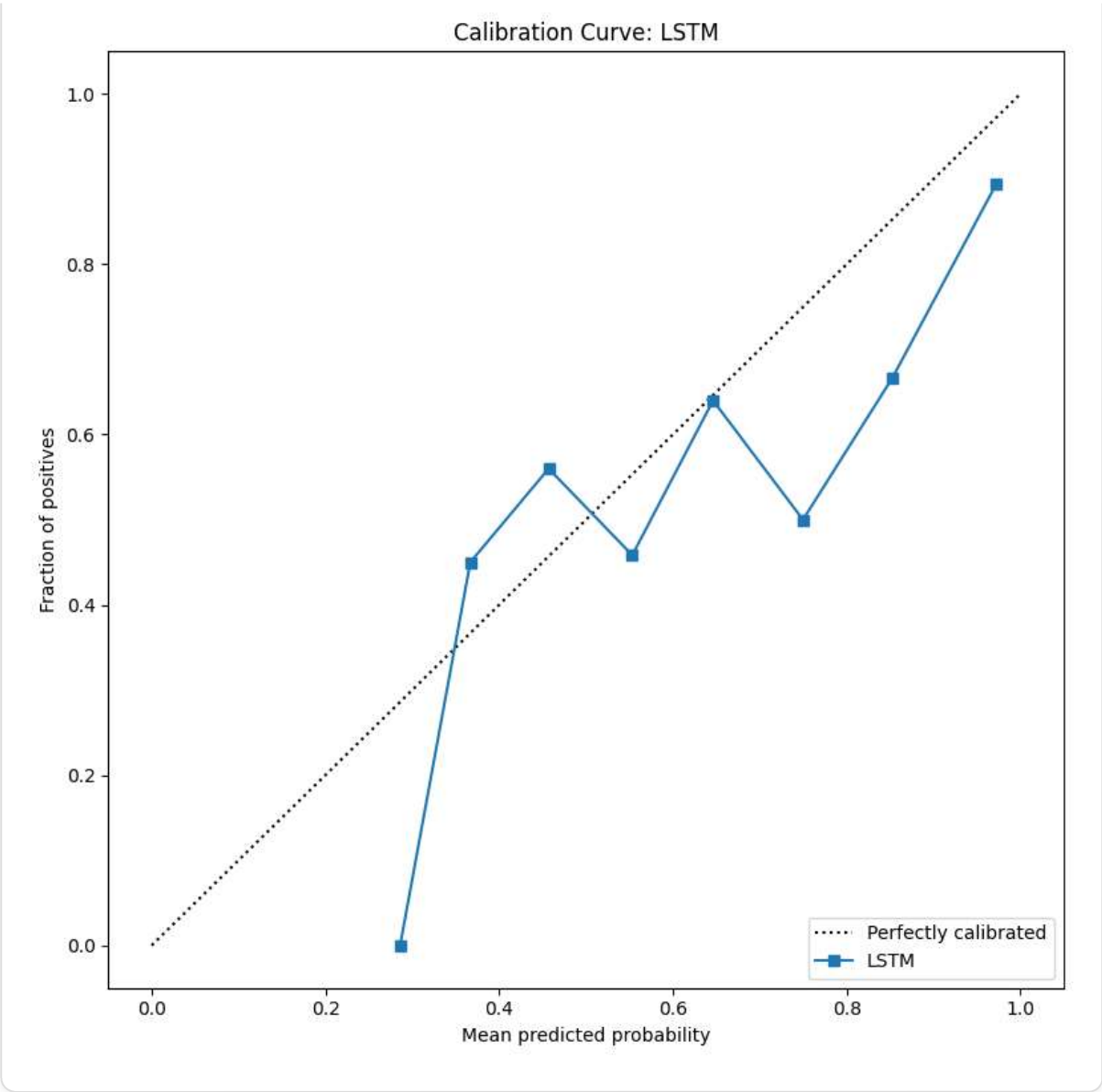


Model: LSTM

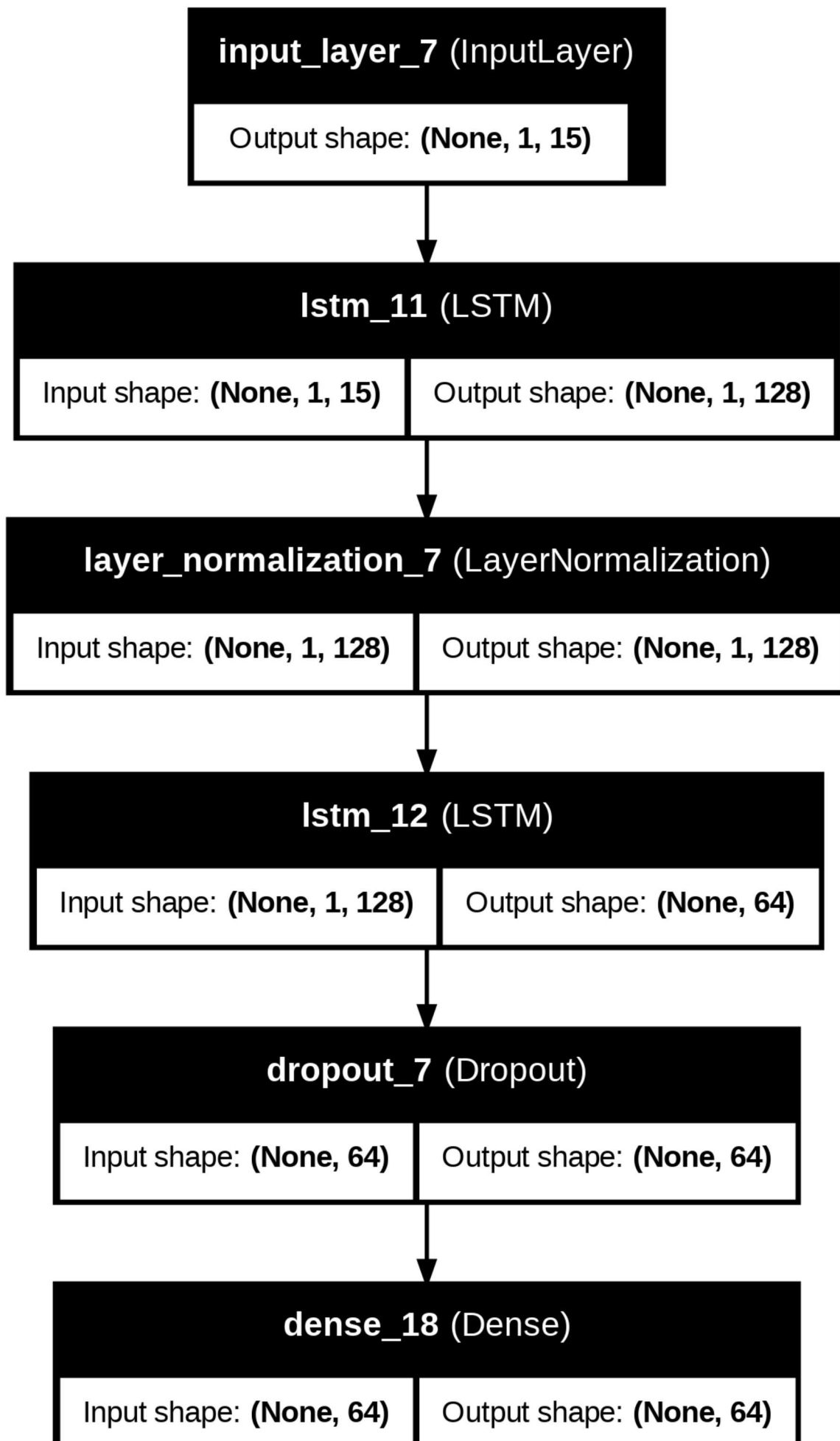
Performance Summary: Acc: 0.5839 | F1: 0.6667 | Prec: 0.6264 | Rec: 0.7125 |
AUC: 0.6340 | MSE: 0.234820

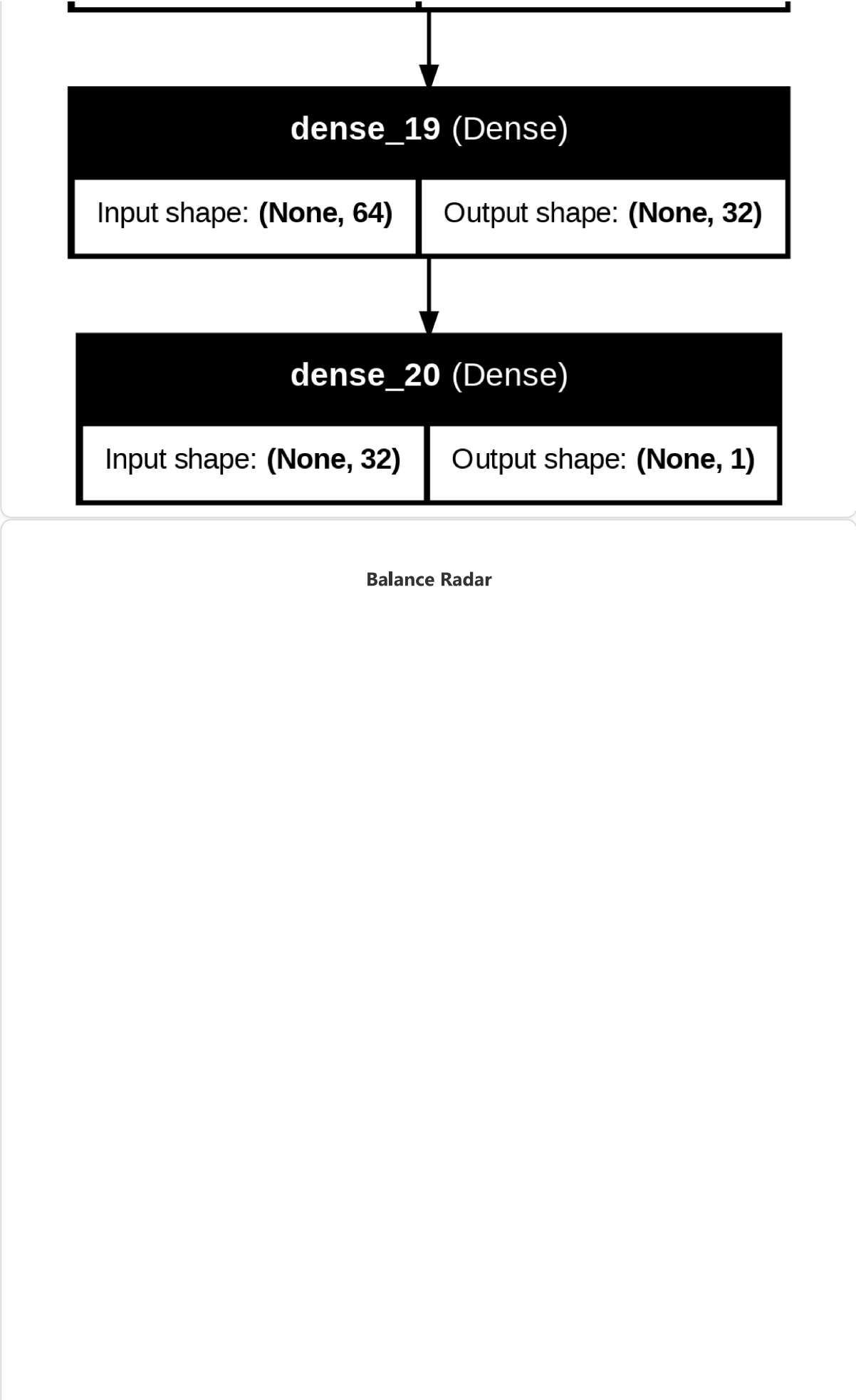
Confusion Matrix



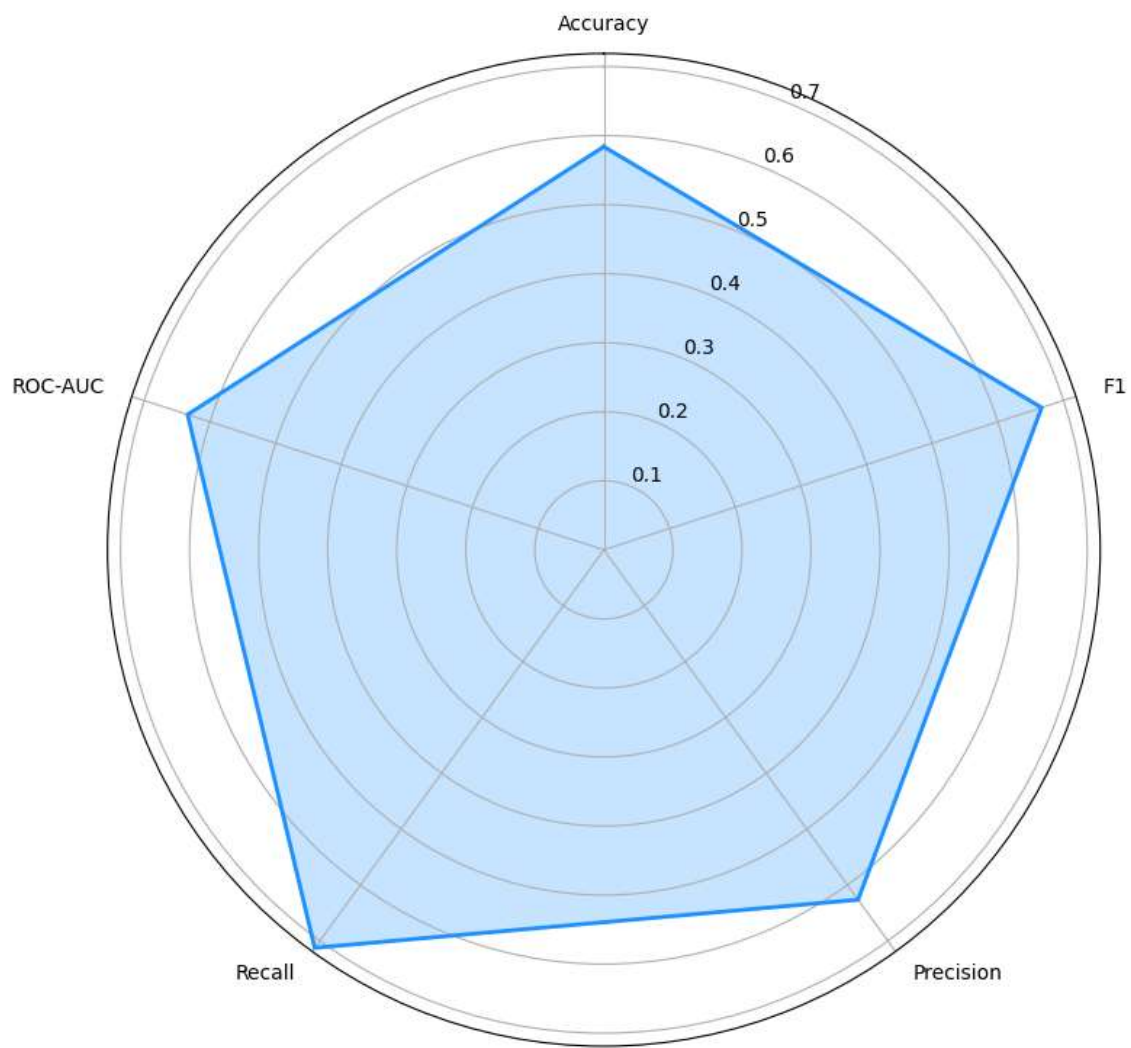


Architecture





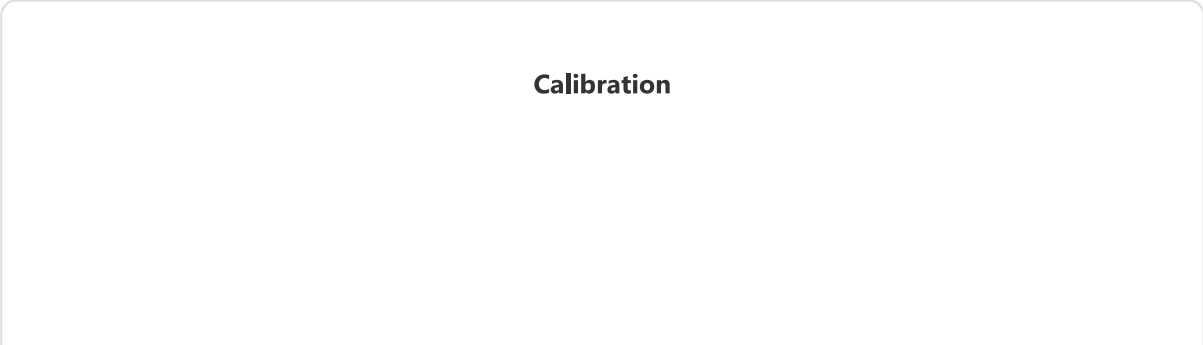
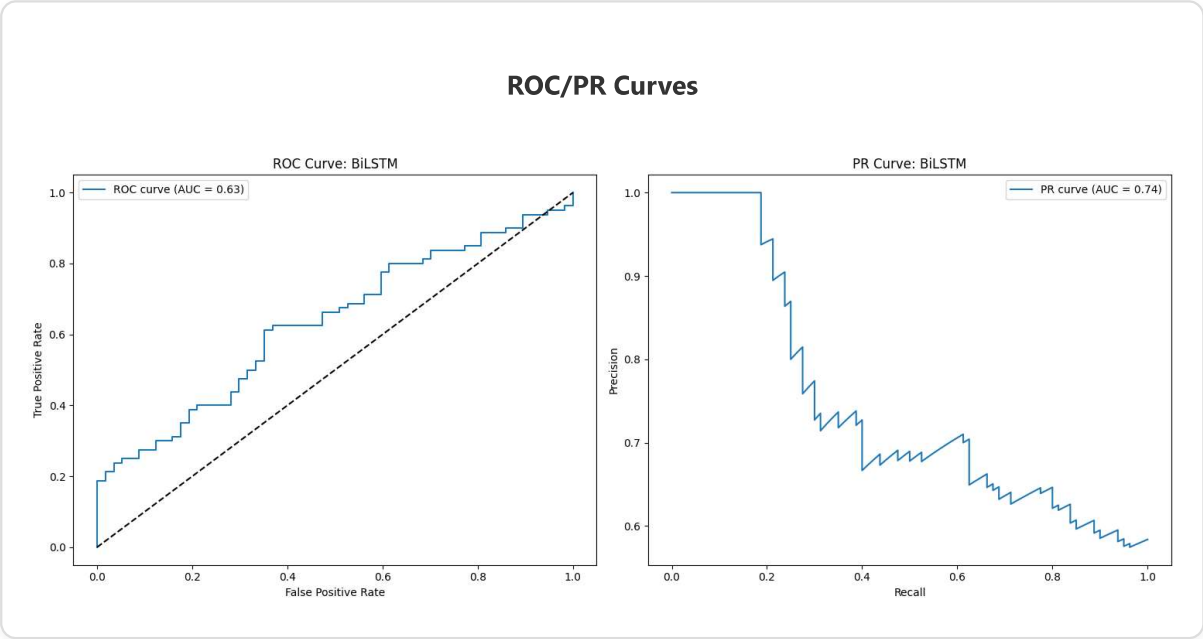
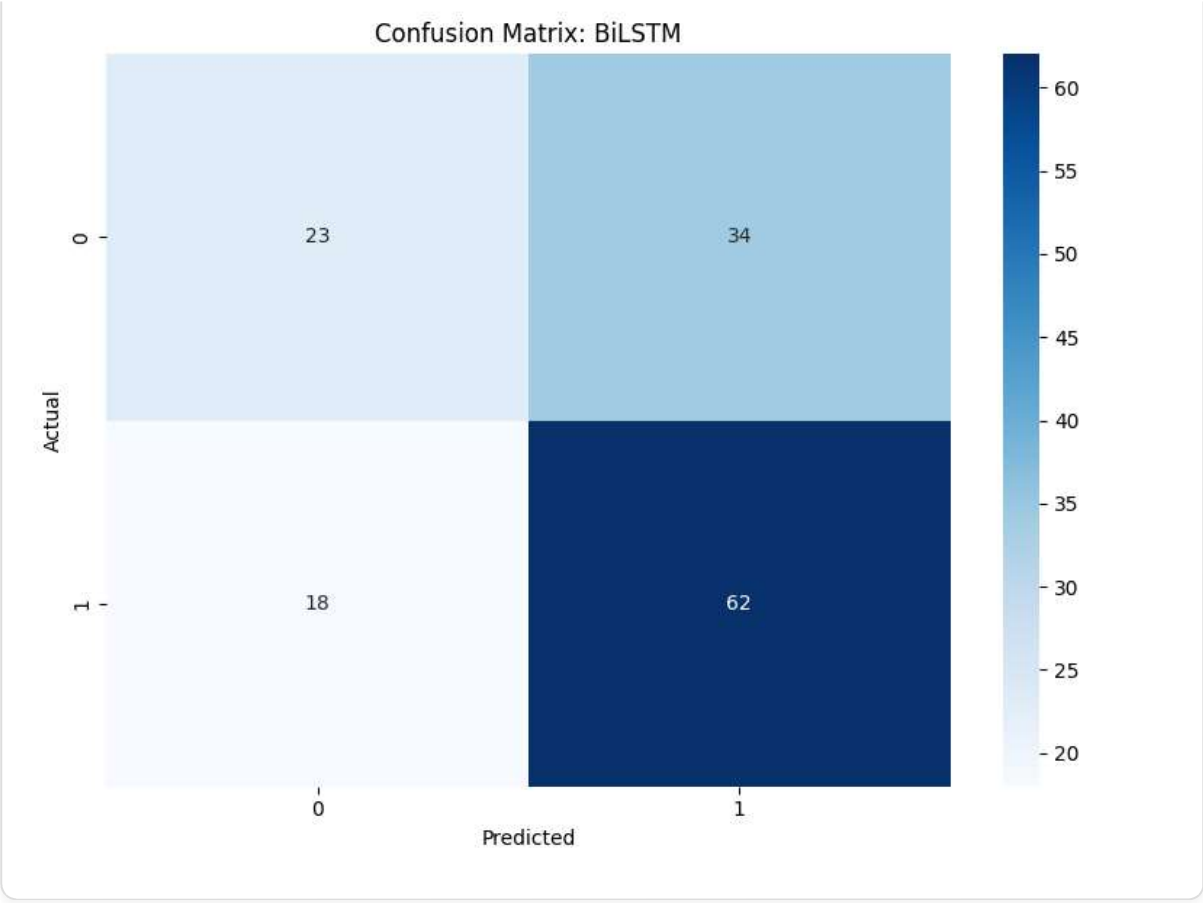
Performance Balance Profile: LSTM

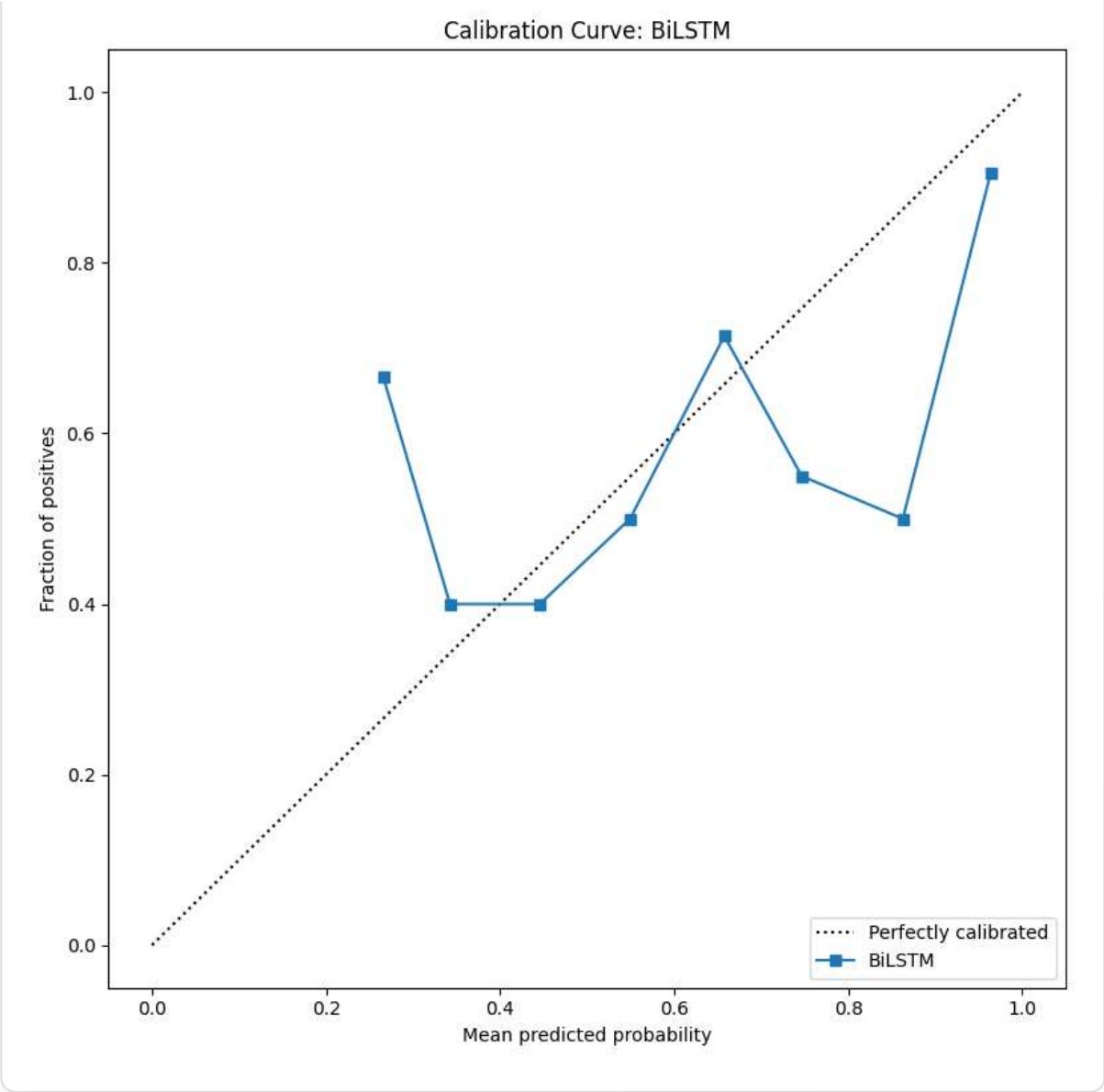


Model: BiLSTM

Performance Summary: Acc: 0.6204 | F1: 0.7045 | Prec: 0.6458 | Rec: 0.7750 |
AUC: 0.6318 | MSE: 0.241692

Confusion Matrix





Architecture

